

Quantum Field Theory – 2
Prof. Sachindeo Vaidya
Centre for High Energy Physics, IISc Bangalore (2023)

Handwritten lecture notes, transcribed

Contents

1	Lecture 1	2
2	Lecture 2	6
3	Lecture 3	10
4	Lecture 4	15
5	Lecture 5	19
6	Lecture 6	24
7	Lecture 7 (30 Jan 2023)	28
8	Lecture 8	31
9	Lecture 9	35
10	Lecture 10–11	39

1 Lecture 1

Goals :—

- QFT 1 recap
 - Scalar & spinor fields
 - L & H formulation
 - canonical quantisation
 - symmetries & conservation laws
 - Interaction & S matrix
 - QED scattering examples
- Functional derivatives / Integrals
- Renormalisation
- Gauge theories (Gauge principle, parallel transport)
- Quantisation
 - Functional Integrals
 - BRST
 - Renormalisation.
- Spontaneous symmetry breaking, Anomalies
- Gauge theories (some non-perturbative questions)
 - Anomalies / Index theories
 - Confinement / Area law
 - t'Hooft & polyakov monopoles

Text books

- (1) Itzykson & Zuber
- (2) Weinberg V II
- (3) Srednicki QFT

(the three above) } Choose any one book.

- Peskin & Schroeder
- Schwartz
- Ramond.

Exams & Assignments –

- Assignments – every week or one per 10 days
 - Exams – Mid term, End term } 4 hrs or more if required.
-

Notations –

$$ds^2 = dt^2 - (d\vec{x})^2$$

Minkowski metric

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

$$p \cdot x = p_\mu x^\mu = p^\mu x_\mu = p_0 x_0 - \vec{p} \cdot \vec{x}$$

$$\epsilon^{ijk} = \epsilon^{123} = \underline{1} \quad \epsilon^{\mu\nu\rho\sigma} = \epsilon^{0123} = 1$$

∂ is sometimes used to denote boundary of space or space-time region. (So, ∂V or $\partial\Sigma$ are boundaries of V & Σ respectively.)

QFT–1 Recap

Particle	Field
(1) spin 0 (uncharged, massive)	scalar field
(2) spin 0 charged boson	complex scalar field
(3) photon (spin –1 massless)	$A_\mu(\vec{x}, t)$ real vector field / potential.
(4) spin $\frac{1}{2}$ fermions ($e^\pm, \gamma$, quarks ...)	spinor field $\Psi_r(\vec{x}, t)$

Scalars –

$$u_k(\vec{x}) = \frac{e^{-ik \cdot x}}{\sqrt{2\omega_k V}} \quad \omega_k = \sqrt{k^2 + m^2} \quad (V \rightarrow \infty \text{ in end})$$

$$\int d^3x \left(u_k^* \left(-i\partial_0 u_{k'} \right) - (i\partial_0 u_k^*) u_{k'} \right) = \delta_{k,k'}$$

where

$$k \equiv (k_1, k_2, k_3) = \frac{2\pi}{L} (n_1, n_2, n_3), \quad V = L^3$$

for $V \rightarrow \infty$, k becomes continuous &

$$\delta_{k,k'} \longrightarrow \frac{(2\pi)^3}{V} \delta(k - k')$$

$$\sum_k \longrightarrow V \int \frac{d^3k}{(2\pi)^3}$$

these wave fun. obey K.G. eqⁿ

$$(\partial^2 + m^2) u_k = 0 \quad \text{i.e.} \quad (\square + m^2) u_k = 0$$

K.G. eqⁿ comes from

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2$$

Fermions –

obey dirac eqⁿ –

$$(-i\gamma^\mu \partial_\mu + m) \Psi(x) = 0 \quad (\text{or}) \quad (i\rlap{\not{\partial}} - m) \psi = 0$$

where

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \mathbb{I}_{4 \times 4}$$

dirac eqⁿ comes from

$$\mathcal{L}_{\text{Dirac}} = \bar{\Psi} (i\rlap{\not{\partial}} - m) \Psi$$

Spinor field ‘ Ψ ’ can be expanded as

$$\Psi_r(x) = \sum_{p,r} \sqrt{\frac{m}{E_p V}} \left(a_{p,r} u_r(p) e^{-ip \cdot x} + b_{p,r}^\dagger v_r(p) e^{ip \cdot x} \right)$$

where u, v are solⁿ of dirac eqⁿ.

we will study about other fields later on.

Functional derivatives –

we know, Action

$$(S) = \int_{\Sigma} d^4x \mathcal{L}(\varphi, \partial_\mu \varphi)$$

$\Sigma \rightsquigarrow$ space time region.

Given a field configⁿ & its derivative we get a real number. So it is a functional.

S = Functional of field & its derivatives.

Note – Thinking of functions as expansion in sine & cosine fun. (Fourier series) is more important in QM & QFT than thinking it in terms of taylor series.

We expand ‘ φ ’ in terms of basis functions $\{f_n\}$

$$\varphi(x) = \sum_n c_n f_n(x)$$

specifying φ is same as specifying c_n .

$$(\varphi + \delta\varphi) \sim \sum (c_n + \delta c_n) f_n(x)$$

we are interested in changing φ in bulk / volumetric region of Σ & keeping φ fixed at surface / boundary represented by $\partial\Sigma$. i.e.

$$\partial\varphi|_{\partial\Sigma} = 0$$

$\partial\Sigma \rightsquigarrow$ represents boundary of region Σ .

$$\delta I = \int_{\Sigma} d^4x \sigma(x) \delta\varphi(x) \quad \left\{ \begin{array}{l} \sigma(x) \text{ is arbitrary fun.} \\ \text{depend upon } \varphi \text{ \& } \partial_\mu \varphi. \end{array} \right.$$

then functional derivative

$$\frac{\delta I}{\delta\varphi(x)} = \sigma(x) \quad (\text{defination})$$

ex :

$$I[\varphi, \partial_\mu \varphi] = \int_{\Sigma} d^4x \left(\frac{1}{2} (\partial_\mu \varphi)(\partial^\mu \varphi) - \frac{1}{2} m^2 \varphi^2 \right)$$

then

$$\frac{\delta I}{\delta \varphi(x)} = -(\partial^2 + m^2) \varphi(x)$$

then we require that,

$$\int_{\Sigma} d^4x \varphi^2(x) < \infty \quad \text{and} \quad \int_{\Sigma} d^4x (\partial \varphi)^2(x) < \infty$$

} To make action finite & finite energy $\int d^3x \mathcal{H} \rightarrow \text{finite}$.

$$\mathcal{F} = \{\text{all } \varphi\text{'s such that } \varphi : \Sigma \rightarrow \mathbb{R} \text{ with above cond}^n\}$$

A Functional (like action S/I) is a real valued function on $\mathcal{F}$ (where $\mathcal{F}$ is infinite dimensional).

Action S or I has physical significance.

if $S \sim \hbar$ use Q.M

if $\frac{S}{\hbar} \gg 1$ we can safely use C.M.

we also expand action as perturbation series of $\hbar$:

$$I = I_0 + \hbar I_1 + \hbar^2 I_2 + \dots$$

$I_0 \downarrow$ classical action; $I_1 \downarrow$ first order quantum correction $\dots$

Canonical quantisation –

(1) Scalar field

For

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \varphi)^2 - \frac{m^2 \varphi^2}{2}$$

we know

$$\begin{aligned} \Pi(\vec{x}, t) &= \partial_0 \varphi(\vec{x}, t) \\ H &= \frac{1}{2} \int d^3x (\Pi^2 + (\nabla \phi)^2 + m^2 \varphi^2) \end{aligned}$$

Imposing canonical commutation relations –

$$[\varphi(\vec{x}, t), \Pi(\vec{x}', t)] = i \delta^3(\vec{x} - \vec{x}')$$

all other equal time commutator vanishes.

(2) Dirac Field –

$$\begin{aligned}\mathcal{L}_0 &= \bar{\Psi} (i\not{\partial} - m) \Psi \\ \Pi &= i\psi^\dagger (\gamma^0)^2 = i\psi^\dagger \quad \dots \quad \{(\gamma^0)^2 = +\mathbb{I}\} \\ H &= \int d^3x \psi^\dagger (i\gamma^0 \gamma^i \partial_i + m\gamma^0) \psi\end{aligned}$$

H is hermitian (we expect it to be).

$$H = \sum_{p,r} E_p \left(a_{p,r}^\dagger a_{p,r} + b_{p,r}^\dagger b_{p,r} \right) - \boxed{2 \sum_p E_p}$$

✓ Becomes infinity. Just throw it.

We can't impose canonical commutation relations, as H is unbounded from below. So we use anti-commutation relations. {also $\Psi(x)$ should be valued in Grassmann no.s}.

Impose anti-commutation relations –

$$\{\Psi_r(\vec{x}, t), \Pi_s(\vec{x}', t)\} = i \delta_{r,s} \delta(\vec{x} - \vec{x}')$$

(Others are zero.)

(No class – Next week)

2 Lecture 2

Propagators –

(1) Free fields

$$\begin{aligned}\varphi(x) &= \int \frac{d^3k}{(2\pi)^3} \left(a_k e^{-ik \cdot x} + a_k^\dagger e^{ik \cdot x} \right) \\ [\varphi(x), \varphi(y)] &= \int d\mu_k \left(e^{-ik \cdot (x-y)} - e^{ik \cdot (x-y)} \right)\end{aligned}$$

↓ a fun. rather than operator

$$\begin{aligned}\int \frac{d^4k}{(2\pi)^4} \frac{-i}{(k^2 - m^2)} e^{-ik \cdot (x-y)} \\ \equiv \Delta(x - y)\end{aligned}$$

since $[\varphi(x), \varphi(y)]$ obeys (Translational Invariance).

Green's fun.

$G(x, y)$ obeys

$$\begin{aligned}(\square_x + m^2) G(x, y) &= -i \delta^4(x - y) \\ G(x, y) &= \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2} e^{-ik \cdot (x-y)}\end{aligned}$$

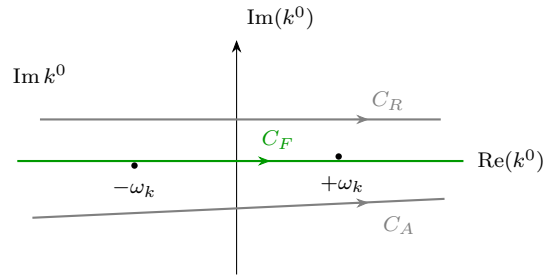
$$\begin{aligned}
& \rightarrow (k^0)^2 - (\vec{k}^2 + m^2) \\
& = \int \frac{dk^0}{2\pi} \int \frac{d^3k}{(2\pi)^3} \frac{i}{(k^0)^2 - (\vec{k}^2 + m^2)} e^{-i(k \cdot (x-y))}
\end{aligned}$$

Choice of contour for k^0 integral is important.

$G_R(x, y)$ = retarded greens fun.

$G_A(x, y)$ = advanced greens fun.

$G_F(x, y)$ = Feynman's Green's fun. (Feynman propagator).



$$G_F(x, y) = \langle 0 | T(\varphi(x)\varphi(y)) | 0 \rangle$$

More generally we are interested in –

$$G(x_1, x_2, x_3 \dots x_n) = \langle 0 | T(\varphi(x_1)\varphi(x_2) \dots \varphi(x_n)) | 0 \rangle$$

we can write generating functional

$$\begin{aligned}
Z[J] &= \sum_N \frac{1}{N!} \int d^4x_1 d^4x_2 \dots d^4x_N G(x_1, x_2 \dots x_N) J(x_1)J(x_2) \dots J(x_N) \\
&= \langle 0 | T e^{\int J(x)\varphi(x)} | 0 \rangle
\end{aligned}$$

J = Source function.

Z obeys the eqⁿ

$$(\partial_x^2 + m^2) \frac{\delta Z[J]}{\delta J(x)} = \langle 0 | T \left[(\partial_x^2 + m^2) \varphi(x) e^{\int J\varphi} \right] | 0 \rangle = -i J(x) Z[J]$$

Free fields –

$$(\partial_x^2 + m^2) \frac{\delta Z_0[J]}{\delta J(x)} = -i J(x) Z_0[J]$$

$$Z_0[J] = \mathcal{N} e^{\frac{1}{2} \int d^4x d^4y J(x) G(x,y) J(y)}$$

Derivation (marginal working)

$$\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - \frac{m^2\phi^2}{2} - \lambda\phi^4 \quad (1)$$

$$= \underbrace{-\frac{1}{2}(\partial\phi)^2 - \frac{m^2\phi^2}{2}}_{\mathcal{L}_0} - J\phi - \lambda\phi^4 \Big|_{J=0}$$

$$\langle 0|0\rangle_J = \int D\phi \exp i \int d^4x (\mathcal{L}_0 - J\phi - \lambda\phi^4)$$

$$\begin{aligned} \mathcal{L} &= -\frac{1}{2}(\partial\phi)^2 - \frac{m^2\phi^2}{2} - J\phi \\ &= \int D\phi \exp i \int d^4x (\mathcal{L}_0 - J\phi) \underbrace{\exp i \int d^4x (-\lambda\phi^4)}_{\text{we will expand}} \\ &= \sum_n \frac{(-i\lambda)^n}{n!} \int D\phi \left(\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right) \int (\phi^4)^n d^4x \end{aligned} \quad (2)$$

let's see.

$$\begin{aligned} \frac{\delta}{\delta J(y)} \left[\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right] &= i \int d^4x \frac{\delta}{\delta J(y)} [\mathcal{L}_0 - J(x)\phi(x)] \times \exp i \int d^4x (\mathcal{L}_0 - J\phi) \\ &= \left[i \int d^4x -\delta(x-y) \phi(x) \right] \times \exp i \int d^4x (\mathcal{L}_0 - J\phi) \\ &= -i \phi(y) \exp i \int d^4x (\mathcal{L}_0 - J\phi) \end{aligned} \quad (3)$$

On Generalizing.

$$\frac{\delta^{4n}}{\delta J(y)^{4n}} \left[\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right] = (-i)^{4n} \phi^{4n}(y) \exp i \int d^4x [\mathcal{L}_0 - J\phi]$$

we got

$$\phi^{4n}(y) \exp i \int d^4x (\mathcal{L}_0 - J\phi) = \frac{\delta^{4n}}{\delta J(y)^{4n}} \left[\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right] \quad (4)$$

use (4) in (2)

$$\begin{aligned} &= \sum_n \frac{(-i\lambda)^n}{n!} \int D\phi \int \frac{\delta^{4n}}{\delta J(y)^{4n}} \left[\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right] \\ &= \sum_n \frac{(-i\lambda)^n}{n!} \int \left(\frac{\delta}{\delta J(y)} \right)^{4n} \int D\phi \exp i \int d^4x (\mathcal{L}_0 - J\phi) \\ &= \sum_n \frac{\left[\int -i\lambda \left(\frac{\delta}{\delta J(y)} \right)^4 \right]^n}{n!} \underbrace{\int D\phi \exp i \int d^4x (\mathcal{L}_0 - J\phi)}_{Z_0(J)} \\ &= \left[\exp \int \left(-i\lambda \frac{\delta^4}{\delta J(y)^4} \right) \right] Z_0(J) \end{aligned}$$

Interactions –

$$\mathcal{L} = \frac{1}{2}(\partial\varphi)^2 - \frac{1}{2}m^2\varphi^2 - \lambda\varphi^4$$

E.O.M :

$$(\partial_x^2 + m^2) \varphi(x) + 4\lambda\varphi^3(x) = 0$$

$$\begin{aligned} Z[J] &= \mathcal{N} e^{-i\lambda \int d^4x \left(\frac{\delta}{\delta J(x)}\right)^4} Z_0[J] \\ &= e^{\frac{1}{2} \int J(x)G(x,y)J(y)} e^{\int J(x)G(x,y)\frac{\delta}{\delta\varphi(y)}} e^{\frac{1}{2} \int G(x,y)\frac{\delta}{\delta\varphi(x)}\frac{\delta}{\delta\varphi(y)}} e^{i\lambda \int \varphi^4(x) d^4x} \end{aligned} \quad (?)$$

where

$$\begin{aligned} V &= \frac{\delta}{\delta\varphi(z_1)} \cdots \frac{\delta}{\delta\varphi(z_n)} \mathcal{F}[\varphi] \quad (?) \\ \mathcal{F}[\varphi] &= \mathcal{N} \exp\left(\frac{1}{2} \int G \frac{\delta}{\delta\varphi(x)} \frac{\delta}{\delta\varphi(y)}\right) \exp(S_{\text{int}}) \quad \left(S_{\text{int}} = \int \lambda\varphi^4\right) \end{aligned}$$

LSZ Formulation –

$$V(x_1 x_2 \cdots x_n) = \prod_j \int i \left(\partial_{x_j}^2 + m^2 \right) G(x_1, x_2 \cdots x_n)$$

Scattering amplitude (S) :

$$\begin{aligned} S(k_1 \cdots k_n \longrightarrow k_{n+1} \cdots k_N) \\ = \prod_{j=1}^n \int u_{k_j}(x_j) i \left(\partial_{x_j}^2 + m^2 \right) \prod_{r=n+1}^N u_{k_r}^*(x_r) i \left(\partial_{x_r}^2 + m^2 \right) G(x_1 x_2 \cdots x_N) \end{aligned}$$

where $x_j = (x_j^0, \vec{x}_j)$.

Electromagnetic field –

$$\begin{aligned} S &= -\frac{1}{4} \int d^4x (F_{\mu\nu} F^{\mu\nu} - A_\mu J^\mu) \\ &= \frac{1}{2} \int d^4x \left[(\vec{E}^2 - \vec{B}^2) - A_0 J_0 + \vec{A} \cdot \vec{J} \right] \end{aligned}$$

where,

$$E_i = F_{0i}, \quad B_i = \frac{1}{2} \epsilon_{ijk} F_{jk}$$

Maxwell's eqⁿ –

$$\begin{aligned} \partial_\mu F^{\mu\nu} &= J^\nu \\ \partial_\mu \left(\frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} \right) &= 0 \end{aligned}$$

A_μ has a gauge symmetry, i.e. $A_\mu \mapsto A_\mu + \partial_\mu \theta(t, x)$ gives same $F_{\mu\nu}$, i.e. ($\vec{E}$ & $\vec{B}$).

redundant eqⁿ –

$$\partial_i E_i = J^0 \quad (\text{Not a time evolution eq}^n).$$

Fix the gauge (or getting rid of redundancies) : one choice is to choose $A^0 = 0$. (A is a 4 vector in other frame A^0 can take non zero value, so there is some tension b/w lorentz invariance & fixing the gauge.)

then split,

$$A_i \equiv A_i^T + A_i^L$$

$A^L \longrightarrow \partial_i f$, and A^T obeys

$$\partial_i A_i^T = 0 \xrightarrow{\text{in } \vec{k} \text{ space}} (\vec{k} \cdot \vec{A}^T = 0)$$

$$E_i = \partial_0 A_i - \partial_i A_0^0 = \partial_0 A_i^T + \partial_i (\partial_0 f)$$

$$\Rightarrow \partial_i E_i = \partial_i \partial_i (\partial_0 f) = J_0$$

$$(\partial_0 f)(x^0, \vec{x}) = \int d^3 y \left[G_c(\vec{x} - \vec{y}) \right] J_0(x^0, \vec{y})$$

$$\downarrow -\frac{1}{4\pi} \frac{1}{|\vec{x} - \vec{y}|}$$

$$S = \frac{1}{2} \int d^4 x [(\partial_0 A_j^T)(\partial_0 A_j^T) - (\partial_i A_j^T)(\partial_i A_j^T)] \\ + \int d^4 x \int d^4 y \frac{1}{2} J_0(x) G_c(x, y) J_0(y) \delta(x^0 - y^0) + \int d^4 x A_i^T J_i$$

The condition $A^0 = 0 = \vec{\nabla} \cdot \vec{A}$ is called radiation gauge. This is action for 2 massless field (the two transverse poles of A_i).

If J^μ is zero, then we have

$$\partial^2 A_i^T = 0$$

plane wave solⁿ e^{-ikx}

it has solutions

$$A_i^T(x) = \sum_{k, \lambda} \left(a_{k\lambda} e_i^{(\lambda)}(k) u_k(x) + c.c \right)$$

$\downarrow$ polarisation vectors

$$\sum_{\lambda=1,2} e_i^{(\lambda)}(k) e_j^{(\lambda)}(k) = \left(\delta_{ij} - \frac{k_i k_j}{\vec{k}^2} \right)$$

There wouldn't be much necessary for the things to come!

3 Lecture 3

Quantisation (QED) :

$$[a_{k\lambda}, a_{k'\lambda'}^\dagger] = i \delta_{kk'} \delta_{\lambda\lambda'} \quad (\hbar = 1)$$

Propagator :

$$\mathcal{D}_{ij}(x, y) = \langle 0 | T (A_i^T(x) A_j^T(y)) | 0 \rangle$$

$$\mathcal{D}_{ij}(x, y) = \int \frac{d^4 k}{(2\pi)^4} \left(\delta_{ij} - \frac{k_i k_j}{\vec{k}^2} \right) \frac{e^{-ik(x-y)}}{k^2 + i\varepsilon}$$

covariant form

$$= \eta_{\mu\nu} \int \frac{d^4 k}{(2\pi)^4} \frac{e^{-ik(x-y)}}{k^2 + i\varepsilon}$$

The S-matrix Functional. :

$$\mathcal{F} = \left(e^{-\frac{1}{2} \int_x \int_y \mathcal{D}_{\mu\nu}(x,y) \frac{\delta}{\delta A_\mu(x)} \frac{\delta}{\delta A_\nu(y)}} \right) e^{i A_\mu J^\mu}$$

$$\{ \mathcal{F}|_{A=0} = Z_0(J) \}$$

Scalar QED

$$\mathcal{L} = (\mathcal{D}_\mu \varphi) (\mathcal{D}^\mu \varphi)^* - m^2 \varphi \varphi^* + \mathcal{L}_{EM}$$

$$\mathcal{D}_\mu = \partial_\mu - iQ A_\mu$$

$$\mathcal{L}_{\text{int}} = -iQ \underbrace{A^\mu (\varphi^* \partial_\mu \varphi - \varphi \partial_\mu \varphi^*)}_{J^\mu} + Q^2 A^\mu A_\mu \varphi^* \varphi$$

The S-matrix Functional:

$$\mathcal{F} = \underbrace{e^{\left(\iint G(x_1, x_2) \frac{\delta}{\delta \varphi(x_1)} \frac{\delta}{\delta \varphi^*(x_2)} \right)}}_{(1)} e^{\left(-\frac{1}{2} \iint \mathcal{D}_{\mu\nu}(x, y) \frac{\delta}{\delta A_\mu(x)} \frac{\delta}{\delta A_\nu(y)} \right)} e^{i S_{\text{int}}}$$

$$S_{\text{int}} = \int d^4x \mathcal{L}_{\text{int}}$$

(1) & $\mathcal{L}_{\text{int}}$ is not gauge invariant but whole expression is gauge invariant.

Spinor QED –

$$\mathcal{L} = \bar{\Psi} (i \not{\partial} - m) \Psi + \mathcal{L}_{EM}$$

$$= \bar{\Psi} (i \not{\partial} - m) \Psi + e \bar{\Psi} \gamma^\mu A_\mu \Psi + \mathcal{L}_{EM}$$

$$\mathcal{F} = \exp \left(-\frac{1}{2} \iint \mathcal{D}_{\mu\nu}(x, y) \frac{\delta}{\delta A_\mu(x)} \frac{\delta}{\delta A_\nu(y)} \right) \cdot \exp \left(- \iint \frac{\delta}{\delta \bar{\Psi}_r(x)} S_{r,s}(x, y) \frac{\delta}{\delta \Psi_s(y)} \right) e^{ie \int A_\mu \bar{\Psi} \gamma^\mu \Psi}$$

Functional integral representation (of $\mathcal{F}$) :

Basis of φ {set of all f^n 's in which we expand φ }

Consider $\varphi(x)$ and $\varphi(x) + \delta\varphi(x)$

$$ds^2 \stackrel{\text{defn}}{=} \|\delta\varphi\|^2 = \int_\Sigma d^4x (\delta\varphi)^2$$

S : action

$$= \sum_n (\delta c_n)^2$$

say ds^2 is metric, where metric depends upon φ .

say $\varphi : \Sigma \longrightarrow S^2$ ($\Sigma \downarrow$ space-time region; $S^2 \rightsquigarrow$ sphere)

if $(\theta, \alpha) \in S^2$ then, φ can be thought of — functions $\theta(x)$ & $\alpha(x)$, ($x \in \Sigma$)

$$\varphi + \delta\varphi \sim (\theta + \delta\theta(x), \alpha(x) + \delta\alpha(x))$$

$$ds^2(\varphi, \varphi + \delta\varphi) = \int_{\Sigma} d^4x \left[(\delta\theta)^2 + (\sin^2 \theta) (\delta\alpha)^2 \right]$$

The set $\{c_n\} \equiv$ local coordinates on $\mathcal{F}$ (set of basis functions).

First consider only n -modes :

$$dV^{(N)} = \sqrt{|\det(g^{(N)})|} dc_1 dc_2 \cdots dc_N$$

$\downarrow N \times N$ matrix

$$I = \int [d\varphi] e^{-\frac{1}{2} \int_x \int_y \varphi(x) M(x,y) \varphi(y)} \quad \text{where} \quad \varphi(x) = \sum c_n f_n(x)$$

$$\text{exponent} = \sum_{mn} c_m M_{mn} c_n$$

$$M_{mn} = \int_{\Sigma} \int_{\Sigma} d^4x d^4y f_m(x) M(x,y) f_n(y)$$

We need M_{mn} to be diagonalizable, $\text{Re}(\text{eigen values}) > 0$.

Consider a function $u_n(x)$ defined by

$$\int d^4y M(x,y) u_n(y) = \lambda_n u_n(x)$$

then write $\varphi \sim \sum a_n u_n$

then,

$$\|\delta\varphi\|^2 = \sum_n (\delta a_n)^2 \quad dV = \prod_n^N da_n$$

$$\begin{aligned} I &= \lim_{N \rightarrow \infty} \int \cdots \int \prod_n^N da_n e^{-\frac{1}{2} \sum \lambda_n a_n^2} \quad (\text{Re}(\lambda) > 0) \\ &= \lim_{N \rightarrow \infty} \prod_{n=1}^N \left(\frac{2\pi}{\lambda_n} \right)^{1/2} \equiv \left(\det \frac{M}{2\pi} \right)^{-1/2} \end{aligned}$$

Truncating to finite N , then taking $N \rightarrow \infty$ is called a Normalisation procedure.

Similarly consider,

$$I[J] = \int [d\varphi] e^{-\frac{1}{2} \int_{\Sigma} \int_{\Sigma} \varphi(x) M(x,y) \varphi(y) + \int_{\Sigma} J(x) \varphi(x)}$$

Completing the square

$$\begin{aligned} I &= \int [d\varphi] e^{-\frac{1}{2} \iint (\varphi - JM^{-1}) M (\varphi - M^{-1}J)} e^{\frac{1}{2} \int JM^{-1}J} \\ &= \left| \frac{M}{2\pi} \right|^{1/2} e^{\frac{1}{2} \iint J(x) M^{-1}(x,y) J(y)} \end{aligned}$$

If φ is complex, then

$$I[J, \bar{J}] = \left(\det \frac{M}{2\pi} \right)^{-1} e^{\iint \bar{J} M^{-1} J}$$

Functional integral repⁿ for fermions –

Grassmann variables (or numbers) are anti-commuting object on which complex conjugate like operation is also defined.

If α, β are grassmann numbers then $\alpha\beta$ & $\beta\alpha$ are also grassmann numbers.

$$\alpha\beta = -\beta\alpha \quad \Rightarrow \quad \{\alpha, \beta\} = 0$$

$$\alpha\beta + \beta\alpha = 0 \quad (\text{Sum of two grassman numbers is zero.})$$

For single grassmann variable η

$$\eta^2 = 0 \quad \text{i.e.} \quad \eta\eta + \eta\eta = 0 \Rightarrow 2\eta^2 = 0 \Rightarrow \eta^2 = 0$$

functions may be expanded as

$$f(\eta) = f_0 \eta^0 + f_1 \eta + \cancel{f_2 \eta^2} + \dots$$

if a function depends upon ordinary variable x & grassmann variable η .

$$f(x, \eta) = f_0(x) \eta^0 + f_1(x) \eta + \cancel{f_2(x) \eta^2} + \dots$$

If a fⁿ depends upon two grassmann numbers & an ordinary variable. (η_1, η_2) .

$$\eta_1^2 = 0 = \eta_2^2 ; \quad (\eta_1 \eta_2 = -\eta_2 \eta_1)$$

$$f(x, \eta_1, \eta_2) = f_0(x) + f_1(x) \eta_1 + f_2(x) \eta_2 + f_{12}(x) \eta_1 \eta_2 + 0$$

$$\bar{\eta} A \eta \quad \left\{ \bar{\eta} = (\bar{\eta}_1, \bar{\eta}_2), \quad \eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \right\}$$

$A \longrightarrow 2 \times 2$ matrix with $\mathbb{R}$ (or $\mathbb{C}$) entries

Lattice discretisation (working pages)

[These pages continue in the notebook after a blank leaf; they form one rough working block on discretising the Klein–Gordon operator on a lattice and are transcribed in the order written.]

$$\begin{aligned} \delta^2 \phi &= \frac{1}{a} \left(\frac{\phi(n + \hat{\mu}) - \phi(n)}{a} - \frac{(\phi(n) - \phi(n - \hat{\mu}))}{a} \right) \\ &= \frac{1}{a^2} (\phi(n + \hat{\mu}) - \phi(n) - \phi(n) + \phi(n - \hat{\mu})) \\ \delta^2 \phi &= \frac{1}{a^2} (\phi(n + \hat{\mu}) + \phi(n - \hat{\mu}) - 2\phi(n)) \end{aligned}$$

with $\phi = e^{ik \cdot x}$, $\phi(n) = e^{ik \cdot n}$:

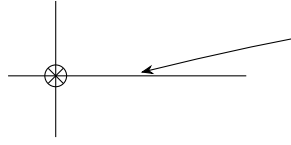
$$\begin{aligned} &= e^{ip \cdot (n + \hat{\mu})} + e^{ip \cdot (n - \hat{\mu})} - 2e^{ip \cdot n} \\ &= e^{ip \cdot n} (e^{ip \cdot \hat{\mu}} + e^{-ip \cdot \hat{\mu}} - 2) \\ &= e^{ip \cdot n} (e^{ip_\mu} + e^{-ip_\mu} - 2) \end{aligned}$$

$$\begin{aligned}
&= e^{ip \cdot n} (2 \cos p_\mu - 2) \\
&= -2e^{ip \cdot n} (1 - \cos p_\mu) \\
&= \left(4e^{ip \cdot n} \sin^2 \frac{p_\mu}{2} + m^2 \right)^{-1} \\
S &= \int \bar{\Psi} M \Psi d^4x
\end{aligned}$$

(margin, in blue:)

$$(\square_x + m^2) G(x, y) = \delta(x - y) ; \quad (\partial_x^2 + m^2) G(x, y) = \delta(x - y)$$

at $p_\mu = 0$,



$$\begin{aligned}
&\frac{1}{\sum_{\mu=1}^4 \sin^2 \frac{p_\mu}{2} + m^2} \\
&\frac{2\pi}{L} (n_1, n_2, n_3) \qquad \frac{2\pi}{L} n_1
\end{aligned}$$

H.W : No doubling in case of scalars, but we see doubling in case of fermions for massless case.

$$(\partial^2 + m^2) \phi(x) \qquad \phi(x) = \langle x | \phi \rangle$$

insert $\mathbb{I}$:

$$\begin{aligned}
&\langle x | \int dp \, |p\rangle \langle p| \phi \rangle \\
&(\partial^2 + m^2) \int dp \, \langle x | p \rangle \hat{\phi}(p) \\
\delta^2 &= \frac{1}{a^2} (\phi(n + \hat{\mu}) + \phi(n - \hat{\mu}) - 2\phi(n))
\end{aligned}$$

$$G(x,y) \,=\, G(|x-y|) \,=\, \int \frac{d^4k}{(2\pi)^4} \, e^{ik \cdot (x-y)} \, \widetilde{G}(k)$$

$\langle k | x - y \rangle$

$$\begin{aligned}
(\square_x + m^2) G(x, y) &= \int (-k^2 + m^2) \frac{d^4k}{(2\pi)^4} e^{ik \cdot (x-y)} \widehat{G}(k) \\
&= \delta^{(4)}(x - y) = \int \frac{d^4k}{(2\pi)^4} e^{ik \cdot (x-y)} \\
(-k^2 + m^2) \widehat{G}(k) &= 1 \quad \Rightarrow \quad \widehat{G}(k) = \frac{1}{-k^2 + m^2}
\end{aligned}$$

$$e^{ik \cdot (x-y)} \qquad \text{with} \quad x = na, \quad y = ma$$

$$\longrightarrow e^{ip \cdot (n-m)}$$

$$\phi(x) = \langle x | \phi \rangle = \int \frac{d^4 p}{(2\pi)^4} \hat{\phi}(p) e^{ip \cdot x}$$

$$\phi(n) = \int \frac{dp}{(2\pi)^4} \hat{\phi}(p) e^{ip \cdot n}$$

$$\phi(n + \hat{\mu}) = \int dp \hat{\phi}(p) \boxed{e^{ip \cdot (n + \hat{\mu})}}$$

$$S[\phi, \partial_\mu \phi] = \int d^4 x \phi \left(\int \frac{d^4 p}{(2\pi)^4} \hat{\phi}(p) \left[e^{ip \cdot (n + \hat{\mu})} + e^{ip \cdot (n - \hat{\mu})} - 2e^{ip \cdot n} \right] \right)$$

$\underbrace{\hspace{1cm}} \downarrow$ Inverse of propagator

$$\mathcal{D}_F(x - y) = \int \frac{d^4 k}{(2\pi)^4} \frac{e^{ik \cdot (x - y)}}{k^2 - m^2}$$

4 Lecture 4

Assignment – 1

- (1) Read chapter 6, 7, 8 — for path integrals
43, 44 — path integral for fermions
- (2) Solve all problems in above chapters.

Functional integrals for fermions :

$$\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \quad a_{ij} \in \mathbb{R} \text{ or } \mathbb{C}$$

$$\bar{\eta} A \eta = (\bar{\eta}_1 \quad \bar{\eta}_2) \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$

$$= (\bar{\eta}_1 \quad \bar{\eta}_2) \begin{pmatrix} a_{11}\eta_1 + a_{12}\eta_2 \\ a_{21}\eta_1 + a_{22}\eta_2 \end{pmatrix}$$

$$\bar{\eta} A \eta = \bar{\eta}_1 a_{11} \eta_1 + \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{21} \eta_1 + \bar{\eta}_2 a_{22} \eta_2$$

$$e^{-\bar{\eta} A \eta} = 1 - \bar{\eta} A \eta + \frac{(\bar{\eta} A \eta)^2}{2!} - \frac{(\bar{\eta} A \eta)^3}{3!} + \dots$$

$$= 1 - \underbrace{(\bar{\eta}_1 a_{11} \eta_1 + \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{21} \eta_1 + \bar{\eta}_2 a_{22} \eta_2)}_{\textcircled{1}}$$

$$\textcircled{2} \left\{ \begin{array}{l} + (\bar{\eta}_1 a_{11} \eta_1 + \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{21} \eta_1 + \bar{\eta}_2 a_{22} \eta_2) \\ \times (\bar{\eta}_1 a_{11} \eta_1 + \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{21} \eta_1 + \bar{\eta}_2 a_{22} \eta_2) \\ + \dots \end{array} \right.$$

$\hookrightarrow$ 2nd term contains $\bar{\eta}_2^2, \bar{\eta}_1^2, \eta_1^2\eta_2^2$ which all vanishes.

② = The terms which survive are

$$\begin{aligned}
& \bar{\eta}_1 a_{11} \eta_1 \bar{\eta}_2 a_{22} \eta_2 + \bar{\eta}_1 a_{12} \eta_2 \bar{\eta}_2 a_{21} \eta_1 \\
& + \bar{\eta}_2 a_{21} \eta_1 \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{22} \eta_2 \bar{\eta}_1 a_{11} \eta_1 \\
= & a_{11} a_{22} \bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2 + a_{12} a_{21} \bar{\eta}_1 \eta_2 \bar{\eta}_2 \eta_1 + a_{21} a_{12} \bar{\eta}_2 \eta_1 \bar{\eta}_1 \eta_2 + a_{22} a_{11} \bar{\eta}_2 \eta_2 \bar{\eta}_1 \eta_1 \\
= & a_{11} a_{22} (\bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2 + \bar{\eta}_2 \eta_2 \bar{\eta}_1 \eta_1) + a_{12} a_{21} (\bar{\eta}_1 \eta_2 \bar{\eta}_2 \eta_1 + \bar{\eta}_2 \eta_1 \bar{\eta}_1 \eta_2) \\
= & \bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2 (a_{11} a_{22} + a_{11} a_{22} + a_{12} a_{21} (-1) + a_{12} a_{21} (-1)) \\
= & \bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2 \cdot 2 \underbrace{(a_{11} a_{22} - a_{12} a_{21})}_{\det A}
\end{aligned}$$

So,

$$e^{-\bar{\eta} A \eta} = 1 - \bar{\eta} A \eta + (\det A) \bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2$$

$$\underline{Z} = 0 \rightsquigarrow \psi(x) \begin{cases} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{cases} \longrightarrow \text{Grassmann co.}$$

$$\psi_1^2 = 0, \quad \underline{\psi_1 \psi_2 = -\psi_2 \psi_1}, \quad \psi_2^2 = 0$$

Recall Z for SHO

$$\begin{aligned}
Z &= \text{Tr} e^{-\beta H} = \sum_{n=0}^{\infty} \langle n | e^{-\beta H} | n \rangle \\
&= \sum_{n=0}^{\infty} e^{-\beta n (1 + \frac{1}{2}) \hbar \omega} = \frac{1}{2 \sinh \frac{\hbar \omega}{2}}
\end{aligned}$$

If we use position basis instead of energy eigenvector basis.

$$\begin{aligned}
Z &= \int dx_0 \langle x_0 | e^{-\beta H} | x_0 \rangle \quad \text{where, } \epsilon = \frac{\beta}{N} \\
&= \int dx_0 \langle x_0 | e^{-\epsilon H} \underbrace{\quad}_I e^{-\epsilon H} \underbrace{\quad}_I \cdots e^{-\epsilon H} | x_0 \rangle
\end{aligned}$$

Inserting $N - 1$ complete set of $|x\rangle$ states

$$\begin{aligned}
I &= \int dx |x\rangle \langle x| \\
&= \int dx_0 \langle x_0 | e^{-\epsilon H} \left| \int dx_1 |x_1\rangle \langle x_1| e^{-\epsilon H} \int dx_2 |x_2\rangle \langle x_2| \cdots \int dx_{N-1} |x_{N-1}\rangle \langle x_{N-1}| x_0 \right\rangle \\
Z &= \int dx_0 dx_1 dx_2 \cdots dx_{N-1} \prod_{i=0}^{N-1} \langle x_i | e^{-\epsilon H} | x_{i+1} \rangle \quad \text{with } (x_N = x_0)
\end{aligned}$$

let us find $\langle x|e^{-\epsilon H}|y\rangle$

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$e^{-\left(\epsilon \frac{p^2}{2m} + \epsilon V(x)\right)} = e^{-\epsilon \frac{p^2}{2m}} e^{-\epsilon V(\hat{x})} + O(\epsilon^2)$$

So,

$$\begin{aligned} \langle x|e^{-\frac{\epsilon p^2}{2m}} e^{-\epsilon V(\hat{x})}|y\rangle & \quad \text{inserting } I = \int \frac{dp}{2\pi} |p\rangle\langle p| \\ &= \int \frac{dp}{2\pi} \langle x|e^{-\frac{\epsilon p^2}{2m}} |p\rangle\langle p|e^{-\epsilon V(\hat{x})}|y\rangle \\ &= \int \frac{dp}{2\pi} e^{-\frac{\epsilon p^2}{2m}} e^{-\epsilon V(x)} e^{ip \cdot (x-y)} \\ &= \frac{1}{(2\pi)^2} e^{-\epsilon V(x)} \int dp e^{-\frac{\epsilon p^2}{2m} + ip \cdot (x-y)} \\ &= \frac{1}{(2\pi)^2} e^{-\epsilon V(x)} \int dp e^{-\frac{\epsilon}{2m} \left(p^2 - \frac{2mi(x-y)}{\epsilon} p + \frac{4m^2(x-y)^2}{\epsilon^2} - \frac{4m^2(x-y)^2}{\epsilon^2} \right)} \\ &= \frac{1}{(2\pi)^2} e^{-\epsilon V(x)} \int dp e^{-\frac{\epsilon}{2m} \left(p - \frac{2mi(x-y)}{\epsilon} \right)^2} e^{-\frac{2m}{\epsilon} (x-y)^2} \\ &= \frac{1}{(2\pi)^2} e^{-(\epsilon V(x) + \frac{2m}{\epsilon} (x-y)^2)} \sqrt{\frac{\pi 2m}{\epsilon}} \end{aligned}$$

So,

$$\begin{aligned} Z &= \int dx_0 dx_1 \cdots dx_{N-1} \prod_{i=0}^{N-1} \frac{1}{(2\pi)^2} e^{-(\epsilon V(x_i) + \frac{2m}{\epsilon} (x_i - x_{i+1})^2)} \cdot \sqrt{\frac{2m\pi}{\epsilon}} \\ &= \frac{\sqrt{m}}{(2\pi)^{3/2}} \left(\frac{1}{\sqrt{\epsilon}} \right)^N \int dx_0 dx_1 \cdots dx_{N-1} e^{-\epsilon \left(\sum_{i=0}^{N-1} \left(V(x_i) + 2m \left(\frac{dx_i}{dt} \right)^2 \right) \right)} \\ & \quad \quad \quad \downarrow S_E \\ &= \sqrt{\frac{m}{\epsilon}} \left(\frac{1}{\sqrt{(2\pi)^3}} \right)^N \int \mathcal{D}x e^{-S_E} \end{aligned}$$

Note : Path integral approach is difficult / lengthy for Q.M. but for field theory this is much helpful approach.

For bosons we have commutator relations but for fermions we have anti commutation relations

$$\begin{aligned} \{a, a^\dagger\} = 1 & \quad \longrightarrow \quad 2a^\dagger a^\dagger = 0, \quad (a^\dagger)^2 = 0 \\ \{a, a\} = \{a^\dagger, a^\dagger\} &= 0 \end{aligned}$$

we also define fermionic vacuum $|0\rangle$ as

$$\begin{aligned} a|0\rangle &= 0 \\ a^\dagger|0\rangle &= \text{some state} \dots |1\rangle \\ (a^\dagger)(a^\dagger)|0\rangle &= 0|0\rangle = 0 \end{aligned}$$

So, $a^\dagger|1\rangle = a^\dagger a^\dagger|0\rangle = 0$

So, we have only two states $|0\rangle$ & $|1\rangle$.

if we try to add one more particle by $(a^\dagger)^2|0\rangle$ which becomes zero. (obeys Pauli-exclusion principle; does not allow 2 particles).

$$H = \frac{\hbar\omega}{2} (a^\dagger a - a a^\dagger) = \hbar\omega \left(a^\dagger a - \frac{1}{2} \right)$$

Such that,

$$H|0\rangle = -\frac{\hbar\omega}{2}|0\rangle$$

$$H|1\rangle = \frac{\hbar\omega}{2}|1\rangle$$

$$\text{So, } H = \frac{\hbar\omega}{2} \sigma_3 = \frac{\hbar\omega}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

we just have 2 eigen states of H .

Partition Fun :

$$\begin{aligned} Z &= \text{Tr} \left(e^{-\beta H} \right) = \underbrace{\langle 0|e^{-\beta H}|0\rangle}_{\textcircled{1}} + \underbrace{\langle 1|e^{-\beta H}|1\rangle}_{\textcircled{2}} \\ &= \langle 0|1 - \beta H + \frac{\beta^2 H^2}{2!} - \frac{\beta^3 H^3}{3!} + \dots|0\rangle + \langle 1|1 - \beta H + \frac{\beta^2 H^2}{2!} - \dots|1\rangle \\ \textcircled{1} &= \langle 0|0\rangle - \beta\langle 0|H|0\rangle + \frac{\beta^2}{2!}\langle 0|H^2|0\rangle - \dots \\ &= 1 + \beta\frac{\hbar\omega}{2} + \frac{\beta^2}{2!}\frac{\hbar^2\omega^2}{4} + \dots = e^{\frac{\beta\hbar\omega}{2}} \end{aligned}$$

Similarly $\textcircled{2} = e^{-\frac{\beta\hbar\omega}{2}}$

$$Z = e^{\frac{\beta\hbar\omega}{2}} + e^{-\frac{\beta\hbar\omega}{2}} = 2 \cosh \frac{\beta\hbar\omega}{2}$$

Path Integrals for fermionic Z :

Grassmann variables c, c^*

Axioms :

(1) c & c^* are independent

(2) $c^2 = (c^*)^2 = 0 \quad \{c, c^*\} = 0$

(3) Integrals –

$$(a) \int dc = 0 = \int dc^*$$

$$(b) \int dc \, c = 1 = \int dc^* \, c^*$$

$$(4) \quad 0 = \{c, dc\} = \{c^*, dc^*\} = \{c, dc^*\} = \{c^*, dc\}$$

$$\Rightarrow \int c dc = - \int dc c = -1$$

$$(5) \quad \{dc, dc\} = \{dc^*, dc^*\} = \{dc, dc^*\} = \{dc^*, dc\} = 0$$

Convention : write '*' variables to extreme left.

$$\int dc^* dc \quad \text{while doing mult}^n \quad - \text{Grassmann variable integrals.}$$

(6) How fermionic operators & Grassmann var. are related –

- 1) c, c^* anticommute with $a, a^\dagger$
- 2) $ca^\dagger$ (or) c^*a behave like ordinary operator
 $ex \quad [ca^\dagger, c^*] = 0$

Note : A state $|c\rangle$ which is labelled by Grassmann variable 'c' is $e^{-ca^\dagger}|0\rangle$

$$e^{-ca^\dagger}|0\rangle = (1 - ca^\dagger)|0\rangle$$

and,

$$\begin{aligned} a|c\rangle &= a e^{-ca^\dagger}|0\rangle \\ &= a (1 - ca^\dagger)|0\rangle \\ &= \cancel{a|0\rangle} \overset{0}{-} a c a^\dagger|0\rangle \\ &= - \underbrace{ac}_{\text{anticommute}} a^\dagger|0\rangle \\ &= + c a a^\dagger|0\rangle \\ a|c\rangle &= c|0\rangle \end{aligned} \quad (1)$$

also

$$\begin{aligned} c|c\rangle &= c (1 - ca^\dagger)|0\rangle \\ &= c|0\rangle - \cancel{c} \overset{0}{c} a^\dagger|0\rangle \\ &= c|0\rangle \end{aligned} \quad (2)$$

(1) & (2) implies $\longrightarrow$

$$\boxed{a|c\rangle = c|c\rangle}$$

5 Lecture 5

We were discussing Grassmann integrals & differentiation.

We discussed that $ca^\dagger, c^*a$ behaves like ordinary operators. Note the order matters.

$$[ca^\dagger, c^*] = 0$$

define $|c\rangle$ & $\langle c|$ as

$$|c\rangle = e^{-ca^\dagger}|0\rangle$$

We saw, $a|c\rangle = c|0\rangle = c|c\rangle$

&

$$\begin{aligned}
\langle c| &= \langle 0|e^{-ac^*} \\
&= \langle 0|(1 - ac^*) \\
&= \langle 0| - \langle 0|a c^* \\
&= \langle 0| - \langle 1|c^* \\
\langle c|a^\dagger &= \langle 0|a^\dagger - \langle 1|c^*a^\dagger \\
&= 0 + \langle 1|a^\dagger c^* \\
&= \langle 0|c^* \quad \text{-- verify!}
\end{aligned}$$

Overlap,

$$\begin{aligned}
\langle c'|c\rangle &= \langle 0|e^{-ac'^*} e^{-ca^\dagger}|0\rangle \\
&= \langle 0|(1 - ac'^*)(1 - ca^\dagger)|0\rangle \\
&= \langle 0|1 - ca^\dagger - ac'^* + ac'^*ca^\dagger|0\rangle \\
&= \langle 0|1 + a^\dagger c + c'^*a - c'^*a c a^\dagger|0\rangle \\
&= \langle 0|0\rangle + \cancel{\langle 0|a^\dagger c|0\rangle} + \cancel{\langle 0|c'^*a|0\rangle} + \langle 0|c'^*c a a^\dagger|0\rangle \\
\langle c'|c\rangle &= 1 + c'^*c \simeq e^{c'^*c}
\end{aligned}$$

Remember coherent states of SHO ($|z\rangle$)

$$I = C \int dz d\bar{z} e^{-\bar{z}z} |z\rangle\langle z|$$

Similarly

$$\begin{aligned}
I &= \int dc^* dc e^{-c^*c} |c\rangle\langle c| \\
&= \int dc^* dc (1 - c^*c) e^{-ca^\dagger}|0\rangle\langle 0|e^{-ac^*} \\
&= \int dc^* dc (1 - c^*c) (1 - ca^\dagger) |1\rangle\langle 0|(1 - ac^*) \\
&= \int dc^* dc (1 - c^*c) (|0\rangle - c|1\rangle) (\langle 0| - \langle 1|c^*) \\
&= \int dc^* dc (1 - c^*c) \left[|0\rangle\langle 0| - c^*|0\rangle\langle 1| - c|0\rangle\langle 0| \right. \\
&\quad \left. + c c^*|0\rangle\langle 1| \right] \\
&= \int dc^* dc (1 - c^*c) |0\rangle\langle 0| - \int dc^* dc (1 - c^*c) c^*|0\rangle\langle 1| \\
&\quad - \int dc^* dc (1 - c^*c) c|0\rangle\langle 0| + \int dc^* dc (1 - c^*c) c c^*|0\rangle\langle 1| \\
&= \cancel{\int dc^* dc |0\rangle\langle 0|} + \int dc^* dc c^*c |0\rangle\langle 0| - \cancel{\int dc^* dc |0\rangle\langle 1|} \\
&\quad + \int dc^* dc e^*c c^*|0\rangle\langle 1| - \cancel{\int dc^* dc c|0\rangle\langle 0|} + \int dc^* dc c c^*|0\rangle\langle 0| \\
&\quad + \int dc^* dc c c^*|0\rangle\langle 1| - \cancel{\int dc^* dc e^*c c c^*|0\rangle\langle 1|}
\end{aligned}$$

$$\begin{cases} c^2 = c^{*2} = 0 \\ \int dc \, c = \int dc^* \, c^* = 1 \\ \{c, c^*\} = 0 \end{cases}$$

We will get,

$$\underline{I} = |0\rangle\langle 0| + |1\rangle\langle 1|$$

Similarly, we can find

$$\begin{aligned} \text{Tr } A &= \langle 0|A|0\rangle + \langle 1|A|1\rangle \\ &= \int dc^* dc \, e^{-c^*c} \langle -c|A|c\rangle \\ &= \langle 0|A|0\rangle + \langle 1|A|1\rangle \end{aligned}$$

Find $Z = \text{Tr } e^{-\beta H}$

$$\begin{aligned} &= \int dc^* dc \, e^{-c^*c} \langle -c|e^{-\beta H}|c\rangle \quad \text{let } \epsilon = \frac{\beta}{N} \\ &= \int dc^* dc \, e^{-c^*c} \langle -c|e^{-\epsilon H} \cdot e^{-\epsilon H} \cdot e^{-\epsilon H} \dots e^{-\epsilon H}|c\rangle \end{aligned}$$

Insert

$$I = \int dc_i^* dc_i \, e^{-c_i^*c_i} |c_i\rangle\langle c_i|$$

We gonna need matrix elements like...

$$\begin{aligned} &e^{-c_{i+1}^*c_{i+1}} \langle c_{i+1}|e^{-\epsilon H(a, a^\dagger)}|c_i\rangle \\ &\simeq e^{-c_{i+1}^*c_{i+1}} \langle c_{i+1}|c_i\rangle e^{-\epsilon H(c_{i+1}^*, c_i)} \quad \left(\longrightarrow e^{c_{i+1}^*c_i}\right) \\ &= \exp \left[-\epsilon \left(c_{i+1}^* \frac{(c_{i+1} - c_i)}{\epsilon} - H(c_{i+1}^*, c_i) \right) \right] \\ Z &= \int dc_N^* dc_N \, dc_{N-1}^* dc_{N-1} \dots dc_1^* dc_1 \, e^{-S_E/\hbar} \end{aligned}$$

where,

$$\begin{aligned} S_E &= \epsilon \sum_{i=1}^N \left[c_{i+1}^* \left(\frac{c_{i+1} - c_i}{\epsilon} \right) + H(c_{i+1}^*, c_i) \right] \\ c_{N+1} &= -c_1 \\ c_{N+1}^* &= -c_1^* \end{aligned}$$

Finally we take $N \rightarrow \infty$, $\epsilon \rightarrow 0$ with $\beta = \epsilon N$

$$Z = \int \mathcal{D}c^*(\tau) \mathcal{D}c(\tau) \, e^{-\int_0^{\beta\hbar} d\tau \, (c^* \dot{c} - H)}$$

$$\left. \begin{aligned} c(\beta\hbar) &= -c(0) \\ c^*(\beta\hbar) &= -c^*(0) \end{aligned} \right\} \begin{array}{l} \text{Antiperiodic boundary cond}^n. \\ \text{in imaginary time axis or} \\ \text{axis. (inv. Temp)} \end{array}$$

Problem :

Consider $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 + gx$

Solve by “completing the square”.

Also do variation of above

$$H = H_{SHO} + \underbrace{g(t)x}_{\text{position}} + \underbrace{pf(t)}_{\text{momentum}}$$

$$H = H_{SHO} + a F^*(t) + a^\dagger F(t) \quad \left(\downarrow \frac{f - ig}{\sqrt{2}} \right)$$

– compute the propagator

– compute the vacuum to vacuum transition amplitude.

$$\langle 0; t_{-\infty} | 0, t_{\infty} \rangle \quad \text{where } \{F, g \rightarrow 0 \text{ as } t \rightarrow \pm\infty\}$$

(margin working)

$$H' = a \left(\frac{f + ig}{\sqrt{2}} \right) + a^\dagger \left(\frac{f - ig}{\sqrt{2}} \right) \quad a = \frac{x + ip}{\sqrt{2}}, \quad a^\dagger = \frac{x - ip}{\sqrt{2}}$$

$$\left(\frac{x + ip}{\sqrt{2}} \right) \left(\frac{f + ig}{\sqrt{2}} \right) + \left(\frac{x - ip}{\sqrt{2}} \right) \left(\frac{f - ig}{\sqrt{2}} \right)$$

$$= \frac{1}{2} (xf + ixg + ipf - pg + xf - ixg - ipf - pg)$$

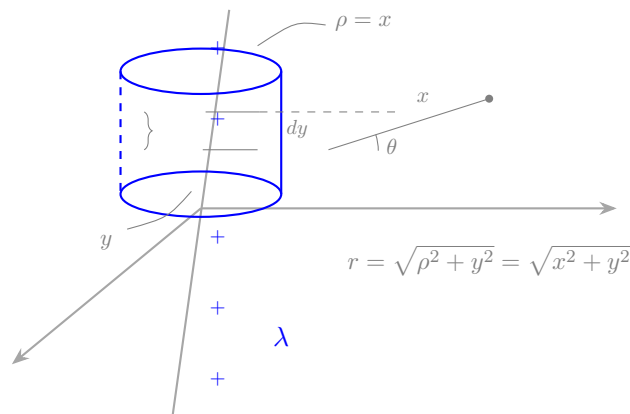
$$= \frac{1}{2} (x(f + ig + f - ig) + p(if - g - if - g))$$

$$= \frac{1}{2} (2xf - 2pg) = \underline{xf} - \underline{pg}.$$

Regularization And Renormalization :

These are not features of QFT alone.

ex : Electrostatics :



$$\phi = \frac{q}{\epsilon_0} \quad \Rightarrow \quad E(2\pi x\ell) = \frac{\lambda\ell}{\epsilon_0}$$

$$E(x) = \frac{\lambda}{2\pi x\epsilon_0}$$

$$\phi(x) - \phi(y) = \int_x^y \vec{E} \cdot \vec{dx}$$

using $E(x)$.

$$\phi(x) - \phi(y) = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{y}{x}$$

But we know

$$\phi(r) = \int \frac{k dq}{r} \quad (\text{here } x = \rho)$$

$$= \frac{1}{4\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{\lambda dy}{\sqrt{x^2 + y^2}}$$

$$\left. \begin{array}{l} y = x \tan \theta \\ dy = x \sec^2 \theta d\theta \end{array} \right\} \Rightarrow \phi(r) = k\lambda \int \frac{x \sec^2 \theta d\theta}{x \sec \theta} = k\lambda \int \sec \theta d\theta$$

$$\Rightarrow \phi(r) = k\lambda \int \frac{\sec \theta (\sec \theta + \tan \theta) d\theta}{\sec \theta + \tan \theta}$$

$$\left. \begin{array}{l} \sec \theta + \tan \theta = t \\ (\sec \theta \tan \theta + \sec^2 \theta) d\theta = dt \end{array} \right\} \quad \phi(r) = k\lambda \int \frac{dt}{t} = k\lambda \ln t = k\lambda \ln (\sec \theta + \tan \theta) \Big|_{-\pi/2}^{\pi/2}$$

$$= \infty - \infty$$

But answer shall be finite!

Similarly when we calculate things in QFT, we sometimes get ∞ , but we know output must be some finite no.

put $\eta = y/x_0$

$$\phi(x) - \phi(x_0) = \frac{\lambda}{4\pi\epsilon_0} \int du \left[\left(\left(\frac{x}{x_0} \right)^2 + u^2 \right)^{-1/2} - (1 + u^2)^{-1/2} \right]$$

use,

$$-\frac{1}{2} \int_1^b d\alpha (u^2 + \alpha)^{-3/2} = (u^2 + b)^{-1/2} - (u^2 + 1)^{-1/2}$$

$$\phi(x) - \phi(x_0) = \frac{-\lambda}{8\pi\epsilon_0} \int_{-\infty}^{\infty} du \int_1^{(x/x_0)^2} d\alpha (u^2 + \alpha)^{-3/2}$$

put,

$$z = u \alpha^{-1/2} (x/x_0)^2$$

$$\begin{aligned} \phi(x) - \phi(x_0) &= -\frac{\lambda}{8\pi\epsilon_0} \int_1^{(x/x_0)^2} \frac{d\alpha}{\alpha} \int_{-\infty}^{\infty} \frac{dz}{(1 + z^2)^{3/2}} \\ &= \frac{-\lambda}{2\pi\epsilon_0} \ln \frac{x}{x_0} \end{aligned}$$

Above way is called dimensional Regularization (DR). DR is faster.

Now we will perform same calculation with DR :

Notice that $\phi(x)$ is scale invariant. $\begin{cases} x \rightarrow kx \\ y \rightarrow ky \end{cases}$

$$\phi(kx) = \phi(x)$$

$$\phi(kx) = \int_{-\infty}^{\infty} \frac{k dy}{k \sqrt{y^2 + x^2}} = \underbrace{\int_{-\infty}^{\infty} \frac{dy}{\sqrt{y^2 + x^2}}}_{\text{This integral is divergent}} = \phi(x)$$

For d -dimension ϕ is not scale invariant!

$$\phi_d(x) = \frac{\lambda}{4\pi\epsilon_0} \mu^{d-1} \int_{-\infty}^{\infty} \frac{d^d y}{\sqrt{y^2 + x^2}}$$

$\downarrow d$ – is dimension of charge confⁿ.

μ has dimension of $\frac{1}{\ell}$ i.e. μ has mass dimension.

Because

$$\phi_d(kx) = k^{d-1} \phi(x)$$

to make it scale invariant we multiply by μ^{d-1} . ($\mu = \text{const.}$)

$$\phi_d(x) = \frac{\lambda}{4\pi\epsilon_0} \mu^{d-1} \int \frac{\rho^{d-1} d\rho}{\sqrt{\rho^2 + x^2}} \int d\Omega_{d-1}$$

i.e. we will have $d - 1$ angular variables & 1 radial variable, in d -dimensions.

$$= \frac{\lambda}{4\pi\epsilon_0} \mu^{d-1} \frac{2(\pi)^{d/2}}{\Gamma(d/2)} \underbrace{\int_0^{\infty} \frac{\rho^{d-1} d\rho}{(\rho^2 + x^2)^{1/2}}}_{I} \quad \text{check!}$$

$$I = \frac{\frac{1}{2} \Gamma(d/2) \Gamma(\frac{1-d}{2})}{\sqrt{\pi}} \quad \leftarrow \text{should include 'x'!}$$

$$(z-1)! = \Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt \quad m! = \Gamma(m+1) = \int_0^{\infty} t^m (e^{-t} dt)$$

6 Lecture 6

Gamma fun. is defined for $\Gamma(x)$, $x > 0$.

for $\Gamma\left(\frac{1-d}{2}\right)$ for $\frac{1-d}{2} < 0$ we get poles.

$$\Gamma_m = (m-1)! = \int_0^{\infty} t^{m-1} e^{-t} dt \quad \leftrightarrow \text{has poles when } m < 0. \quad \text{Similarly } \Gamma\left(\frac{1-d}{2}\right) \text{ has poles for } \frac{1-d}{2} < 1 \Rightarrow 1-d < 2 \Rightarrow d > -1$$

↑ we are extending domain of d from +ve integer to real numbers, by analytic continuation.

Expand ϕ_d in powers of $1 - d \equiv \epsilon$

$$\begin{aligned} \text{if } \frac{1-d}{2} &= -\text{ve integer} \\ 1-d &= -\text{ve even integer} \\ \Rightarrow d &= +\text{ve odd integer} \end{aligned}$$

Note that we can't add quantities of different dimensions.

So $\exp(\text{dimensionful quantity})$ = does not make sense.

μ^{d-1}, x^{d-1} if expanded separately then above problem arise.

So, we combine

$$\begin{aligned} (\mu x)^{d-1} &= (\mu x)^{-\epsilon} \\ &= 1 - \epsilon \ln \mu x + O(\epsilon^2) \quad > \text{should be minus!} \end{aligned}$$

$$x^n = e^{\ln x^n}; \quad (\mu x)^{-\epsilon} = e^{\ln(\mu x)^{-\epsilon}} = 1 + \ln(\mu x)^{-\epsilon} + \frac{(\ln(\mu x)^{-\epsilon})^2}{2!} + \dots = 1 - \epsilon \ln \mu x + O(\epsilon)^2$$

$$\pi^{\frac{d-1}{2}} = 1 - \frac{\epsilon}{2} \ln \pi + O(\epsilon^2)$$

$$\Gamma\left(\frac{1-d}{2}\right) = \Gamma(-\epsilon/2) = \frac{-2}{\epsilon} \Gamma(1 - \epsilon/2) \quad (?)$$

$$\Gamma(z+1) = z \Gamma(z); \quad \Gamma(-\epsilon/2) = \left(-\frac{\epsilon}{2}\right)^{-1} \Gamma\left(-\frac{\epsilon}{2} + 1\right) = \left(-\frac{2}{\epsilon}\right) \Gamma\left(1 - \frac{\epsilon}{2}\right)$$

$$= \frac{-2}{\epsilon} \left(1 + \frac{\epsilon \gamma}{2} + O(\epsilon^2) + \dots\right)$$

So,

$$\phi_d(x) = \frac{-\lambda}{2\pi\epsilon_0} \left[\frac{1}{\epsilon} + \underbrace{\frac{1}{2} \ln \pi + \frac{\gamma}{2} + \ln \mu x}_{\ln \mu' x} + O(\epsilon) \right]$$

↓ Can be dropped for $d = 1 \Rightarrow \underline{\epsilon \equiv 0}$.

define

$$\phi_{d=1}^d(x) = \frac{-\lambda}{2\pi\epsilon_0} (\ln \mu' x)$$

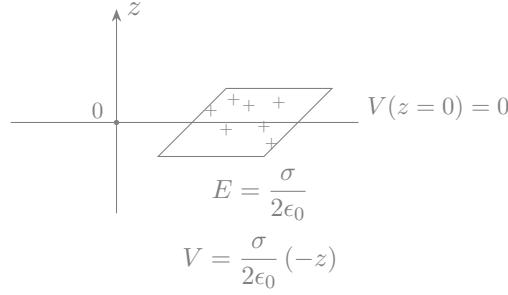
$$\phi^R(x) = \frac{A}{\epsilon} + \left(\frac{-\lambda}{2\pi\epsilon_0}\right) \ln \mu' x$$

$$\phi^R(y) = \frac{A}{\epsilon} + \left(\frac{-\lambda}{2\pi\epsilon_0}\right) \ln \mu' x'$$

$$\phi^R(x) - \phi^R(x') = \frac{-\lambda}{2\pi\epsilon_0} \ln\left(\frac{x}{x'}\right) \quad \leftarrow \text{It is potential difference that are finite!}$$

We have found this answer from basic defⁿ of potential $V(r) = \int \frac{k dq}{r}$

For 2d charge confⁿ.



Generating Functional for interacting scalar field :

$$Z[J] = e^{\frac{1}{2}\langle JGJ \rangle} \left[e^{\langle JG\delta \rangle} \mathcal{F}[\varphi] \right]_{\varphi=0}$$

Where,

$$\langle JGJ \rangle = \int d^4x d^4y J(x) G(x, y) J(y) \quad \text{Check } \varphi^4 d^4x ?$$

$$\mathcal{F}[\varphi] = N e^{\frac{1}{2}\langle \delta G \delta \rangle} e^{-\frac{i\lambda}{4!} \int \varphi^4 dx}$$

$$Z_0[J] = e^{\frac{1}{2}\langle JGJ \rangle}$$

So,

$$Z[J] = N \left[e^{\frac{i\lambda}{4!} \int d^4x \left(\frac{\delta}{\delta J(x)} \right)^4} \right] Z_0[J]$$

Gaussian functional integral

↓

Greens fun are functional integrals

↓

$Z_0[J]$ as a functional integral

↓

Z as a functional integral.

$$Z[J] = N \int [d\varphi] e^{iS - \epsilon \int \varphi^2 + J\varphi}$$

$$S \rightarrow S_0 - \frac{\lambda}{4!} \int d^4x \varphi^4(x) \quad \begin{cases} S = S_M \\ S_E = -iS_M \end{cases}$$

The gaussian integral is of type

$$\int dx e^{i\alpha x^2 - \epsilon x^2} = \sqrt{\frac{\pi}{\epsilon - i\alpha}}$$

We can equally work with euclidean action.

$$t \rightarrow -i\tau, \quad S_E = -iS_M \quad \text{here, } \tau = \text{imag}, t = \text{real}$$

$$Z[J] \sim \int [d\varphi_E] e^{-S_E}$$

$$S_E = \int d^4x \left(\frac{(\partial\varphi)^2}{2} + \frac{\mu^2\varphi^2}{2} + \frac{\lambda\varphi^4}{4} \right)$$

Renormalisation (why in QFT) :

1. loop integrals involve integrals over arbitrary large momenta.
2. Theory (i.e. our $\mathcal{L}$) is not valid at extremely high energies / momenta. (We impose a cut off on momenta, then integrals will be dependent on momentum cut off (say Λ). (known as Λ -sensitive integrals for a given diagram.)
 $\Rightarrow$ Then loop integrals become finite but depend on Λ .

This is like cutting infinite wire to make it wire of finite length ℓ .

where, $\ell \rightarrow \infty$ is approximation of real world.

$$\ell [\underline{V \sim \ln r} \quad \longrightarrow \quad r' \quad \left(V(r') \neq \sim \ln(r) \sim \frac{1}{r} \right)$$

Transition from $\ln(r) \longrightarrow \frac{1}{r}$.

3. Scattering amplitudes depend on parameters of $\mathcal{L}$ (like $\lambda_0, \mu_0 \dots$) $\leftrightarrow$ Bare Coupling const. & the actual measured couplings $(\lambda, \mu \dots)$ are functions of λ_0, μ_0 and Λ (momentum cut off).

$$\lambda = f(\lambda_0, \mu_0, \Lambda) ; \quad \mu = g(\lambda_0, \mu_0, \Lambda)$$

What ? { The Λ -dependence of calculated results can be absorbed between μ_0, λ_0 and $\mu, \lambda \dots$

$\longrightarrow$ let us see this for φ^4 theory.

Start with,

$$\begin{aligned} \mathcal{L}_M &= \frac{1}{2} \alpha^2 (\partial\varphi)^2 - \frac{\mu^2 \varphi^2}{2} - \frac{\lambda}{4} \varphi^4 - J\varphi \\ \mathcal{L}_E &= \frac{1}{2} \alpha^2 (\partial\varphi)^2 + \frac{\mu^2 \varphi^2}{2} + \frac{\lambda}{4} \varphi^4 + J\varphi \end{aligned}$$

So,

$$Z[J] = \int [d\varphi] e^{-S_E + J\varphi} \quad (\text{Meaning of source term ?})$$

$\downarrow$ generates correlation functions / greens fun. which gives scattering amplitudes.

Correlation functions

$$G^{(n)}(x_1, x_2 \dots x_n; J) = \frac{1}{Z[J]} \frac{\delta}{\delta J(x_1)} \frac{\delta}{\delta J(x_2)} \dots \frac{\delta}{\delta J(x_n)} Z[J]$$

$\longleftarrow$ includes disconnected diagram as well

Connected correlation functions. (How ?) $\rightarrow$ will derive tomorrow!

$$G_c^{(n)}(x_1, x_2 \dots x_n; J) = \frac{\delta}{\delta J(x_1)} \dots \frac{\delta}{\delta J(x_n)} \ln Z[J]$$

Mass Renormalisation :

(Srednicki S.24)

Quantity of interest is 2-point vertex function

$$\begin{aligned}\Gamma^{(2)}(k) &= \left(\text{---} \xrightarrow{k} \text{---} \right)^{-1} + \frac{\text{---} \bigcirc^q \text{---}}{k} + O(\lambda^2) \\ &= (\alpha^2 k^2 + \mu^2) + \frac{\lambda}{2} \int^\Lambda \frac{d^4 q}{(\alpha^2 q^2 + \mu^2) (2\pi)^4} + O(\lambda^2) \quad - \quad \textcircled{1}\end{aligned}$$

$\Lambda \leftarrow$ cut off

We demand that, $\Gamma^{(2)}(k=0)$ be equal to (physical mass)².

$$\begin{aligned}\Gamma^{(2)}(k) \Big|_{k=0} &= m^2 \quad \left(\tilde{q} = \frac{q}{2\pi} \right) \\ \Gamma^{(2)}(0) &= m^2 = \alpha^2(0) + \mu^2 + \frac{\lambda}{2} \int_0^\Lambda \frac{d^4 \tilde{q}}{(\alpha^2 q^2 + \mu^2)} + O(\lambda^2) \\ \Gamma^{(2)}(k) &= \Gamma^{(2)}(k) + 0 \\ &= \Gamma^{(2)}(k) - \Gamma^{(2)}(0) + m^2 \\ &= \alpha^2 k^2 + \mu^2 + \frac{\lambda}{2} \int_0^\Lambda \frac{\cancel{d^4 q}}{(2\pi)^4} \frac{1}{(\alpha^2 q^2 + \mu^2)} + O(\lambda^2) \\ &\quad - \left(0 + \mu^2 + \frac{\lambda}{2} \int_0^\Lambda \frac{\cancel{d^4 \tilde{q}}}{(\alpha^2 \tilde{q}^2 + \mu^2)} + O(\lambda^2) \right) \\ &\quad + m^2 \\ \boxed{\Gamma^{(2)}(k) &= \alpha^2 k^2 + m^2 + O(\lambda^2)}\end{aligned}$$

7 Lecture 7 (30 Jan 2023)

Reading Assignment – 2 Read Srednicki chapters 12–20. and solve all problems.

Mass Renormalisation –

$\Gamma^{(2)}(k)$ depends on Λ only then $\exists$ relation b/w μ & m .

$$\int \frac{d^4 \tilde{q}}{\alpha^2 q^2 + \mu^2} = \frac{V(S^{d-1})}{(2\pi)^d} \int_0^\Lambda \frac{q^{d-1} dq}{\alpha^2 q^2 + \mu^2} ; \quad \text{where } \left\{ d^d \tilde{q} = \frac{d^d q}{(2\pi)^d} \right.$$

For $d = 4$;

$$\frac{V(S^3)}{(2\pi)^4} \int_0^\Lambda \frac{q^3 dq}{\alpha^2 q^2 + \mu^2} = \left(\alpha^2 \Lambda^2 + \mu^2 \ln \mu^2 - \underbrace{\mu^2 \ln (\mu^2 + \alpha^2 \Lambda^2)}_{\downarrow} \right)$$

Subleading term.

(margin working)

$$\begin{aligned}
\int \frac{dq \, q^3}{\alpha^2 q^2 + \mu^2} &= \frac{1}{\alpha^2} \int_0^\Lambda \frac{dq (\alpha^2 q^2 + \mu^2 - \mu^2) q}{\alpha^2 q^2 + \mu^2} & \alpha^2 q^2 + \mu^2 &= t \\
&= \frac{1}{2\alpha^4} \int \frac{dt (t - \mu^2)}{t} & 2\alpha^2 q \, dq &= dt \\
&= \frac{1}{2\alpha^4} \int dt \left(1 - \frac{\mu^2}{t} \right) = \frac{1}{2\alpha^4} (t - \mu^2 \ln t) \\
&= \frac{1}{2\alpha^4} (\alpha^2 q^2 + \mu^2 - \mu^2 \ln (\alpha^2 q^2 + \mu^2)) \Big|_0^\Lambda \\
&= \frac{1}{2\alpha^4} (\alpha^2 \Lambda^2 + \mu^2 - \mu^2 \ln (\alpha^2 \Lambda^2 + \mu^2) - (\mu^2 - \mu^2 \ln \mu^2)) \\
&= \frac{1}{2\alpha^4} \left(\alpha^2 \Lambda^2 + \cancel{\mu^2} - \mu^2 \ln \left(\frac{\alpha^2 \Lambda^2 + \mu^2}{\mu^2} \right) \right) \\
&= \frac{1}{2\alpha^4} \left(\alpha^2 \Lambda^2 - \mu^2 \ln \left(\frac{\alpha^2 \Lambda^2}{\mu^2} + 1 \right) \right)
\end{aligned}$$

From eqⁿ ① of 6th lecture –

$$m^2 = \mu^2 + \frac{\lambda}{2} \left(\alpha^2 \Lambda^2 + \mu^2 \ln \left(\frac{\mu^2}{\mu^2 + \alpha^2 \Lambda^2} \right) \right)$$

$$\text{This gives } \mu^2 = \mu^2(\Lambda; m^2)$$

look at 2 loops –

$$\begin{aligned}
\Gamma^{(2)} &= (\text{tree level} + 1 \text{ loop}) + \text{diagram with two bubbles} \\
&+ \text{diagram with a bubble}
\end{aligned}$$

$$\begin{aligned}
&= \mu^2 + \alpha^2 k^2 + \frac{1}{2} \int \frac{d^d \tilde{q}}{\alpha^2 q^2 + \mu^2} - \lambda^2 \left(\frac{1}{4} \int \frac{d^d \tilde{q}_1 \, d^d \tilde{q}_2}{(\alpha^2 q_1^2 + \mu^2) (\alpha^2 q_2^2 + \mu^2)} \right. \\
&\quad \left. + \frac{1}{6} \int \frac{d^d \tilde{q}_1 \, d^d \tilde{q}_2}{(\alpha^2 q_1^2 + \mu^2) (\alpha^2 q_2^2 + \mu^2) (\alpha^2 (k - Q)^2 + \mu^2)} \right) + O(\lambda^3)
\end{aligned}$$

Again define $\Gamma^{(2)}(k=0) = m^2$ and write

$$\Gamma^{(2)}(k) = \Gamma^{(2)}(k) - \Gamma^{(2)}(0) + m^2 + O(\lambda^3)$$

$$\Gamma^{(2)}(k) = \alpha^2 k^2 + m^2 - \frac{\lambda^2}{6} \Delta A(k)$$

$$\Delta A(k) = A(k) - A(0) \quad Q = \tilde{q}_1 + \tilde{q}_2$$

$$\begin{aligned}\Delta A(k) &= \int \frac{d^d \tilde{q}_1 d^d \tilde{q}_2}{(\alpha^2 q_1^2 + \mu^2)(\alpha^2 q_2^2 + \mu^2)} \left(\frac{1}{\alpha^2 (k-Q)^2 + \mu^2} - \frac{1}{\alpha^2 Q^2 + \mu^2} \right) \\ &= \int \frac{d^d \tilde{q}_1 d^d \tilde{q}_2}{(\alpha^2 q_1^2 + \mu^2)(\alpha^2 q_2^2 + \mu^2)} \left(\frac{\alpha^2 Q^2 - \alpha^2 (k-Q)^2}{(\alpha^2 (k-Q)^2 + \mu^2)(\alpha^2 Q^2 + \mu^2)} \right)\end{aligned}$$

It is divergent for $(2d - 4 + 1 - 4 = 2d - 7)$ (?)

λ should be small so that, it nullifies effect of large Λ when series expanded in λ . Why so much small ?

Eliminate μ in favour of $m \rightarrow$ replace $\mu \leftrightarrow m$ in RHS; (?)

$$\begin{aligned}\Gamma^{(2)}(k) &= m^2 + \alpha^2 k^2 - \frac{\lambda^2}{6} \Delta A(k) \\ \mu^2 &= m^2 - \frac{\lambda}{2} \int \frac{d^d \tilde{q}}{\alpha^2 q^2 + m^2} + \lambda^2 \left[\int \frac{d^d \tilde{q}_1 d^d \tilde{q}_2}{(\alpha_1^2 q_1^2 + m^2)(\alpha_2^2 q_2^2 + m^2)} + \frac{1}{6} A(0) \right] + O(\lambda^3)\end{aligned}$$

We have one more relation available to re-define

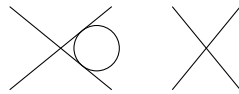
$$\begin{aligned}a^2 &\equiv \left. \frac{d\Gamma^{(2)}(k)}{dk^2} \right|_{k=\lambda} \\ \Gamma^{(2)}(k) &= \Gamma^{(2)}(k) - k^2 \left. \frac{d\Gamma^{(2)}}{dk^2} \right|_k + k^2 \left. \frac{d\Gamma^{(2)}}{dk^2} \right|_k \\ &= m^2 + \alpha^2 k^2 - \frac{\lambda^2}{6} (\Delta A(k)) - k^2 \left. \frac{d\Gamma^{(2)}}{dk^2} \right|_k \\ &\quad + k^2 a^2 + O(\lambda^2) \\ &= m^2 + k^2 \left(a^2 - \frac{\lambda^2}{6} B(k, \lambda) \right) + O(\lambda^3) \\ \downarrow &\quad \frac{\Delta A(k)}{k^2} + \frac{6}{\lambda^2} \left(\left. \frac{d\Gamma^{(2)}}{dk^2} \right)_k \right) - \alpha^2\end{aligned}$$

So we know B ;

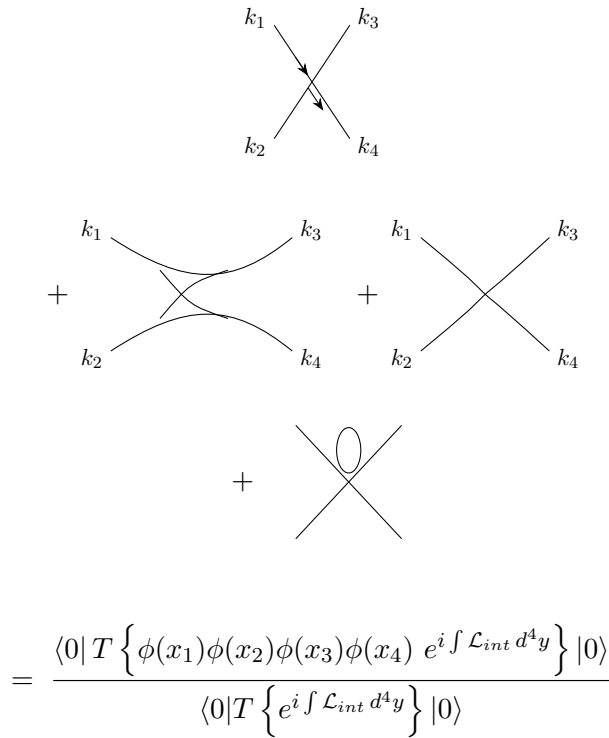
$$\frac{d\Gamma^{(2)}}{dk^2} = \text{[?]} = -\alpha \quad (\text{in free theory})$$

Renormalisation of λ :-

Eliminate λ in favour of $g \equiv \Gamma^{(4)}(0, 0, 0)$



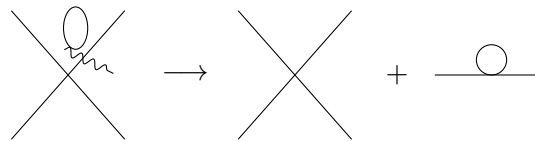
Proceed as earlier :-



$$= \frac{\langle 0 | T \left\{ \phi(x_1) \phi(x_2) \phi(x_3) \phi(x_4) e^{i \int \mathcal{L}_{int} d^4 y} \right\} | 0 \rangle}{\langle 0 | T \left\{ e^{i \int \mathcal{L}_{int} d^4 y} \right\} | 0 \rangle}$$

disconnected diagrams cancels away because of $\log[Z]$ term in path integral.

Path integrals shouldn't be worried about because they fall apart into 2 diagrams.



$$\text{Diagram with loop on contact} \rightarrow \text{Contact diagram} + \text{Diagram with loop on external line}$$

$$\Gamma^{(2)}(k_1, k_2, k_3) = \lambda - \frac{\lambda^2}{2} \int \frac{d^d \tilde{q}}{\alpha^2 q^2 + m^2} \left(\frac{1}{a^2 (k - q)^2 + m^2} + \frac{1}{a^2 (k_1 + k_3 - q)^2 + m^2} + \frac{1}{a^2 (k_1 + k_3 - q)^2 + m^2} \right)$$

Add and subtract $g = \Gamma^{(4)}(0, 0, 0)$

$$\Gamma^{(4)}(k_1, k_2, k_3) = \Gamma^{(4)}(k_1, k_2, k_3) - \Gamma^{(4)}(0, 0, 0)$$

8 Lecture 8

Infrared cut off — means low energy cut off
ultraviolet cut off means high " " off.

$$m^2 = \mu^2 + \frac{1}{2} \left(\alpha^2 \Lambda^2 + \mu^2 \ln \frac{\mu^2}{\mu^2 + \alpha^2 \Lambda^2} \right)$$

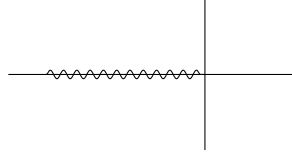
$$Z(m, \lambda) = \int [d\phi] e^{-\frac{1}{2} \int d^4 x ((\partial\phi)^2 + m^2 \phi^2 + \lambda \phi^4)}$$

$$= Z_0(m, 0) + \lambda Z_1(\quad) + \lambda^2 Z_2(\quad) + \dots$$

We know,

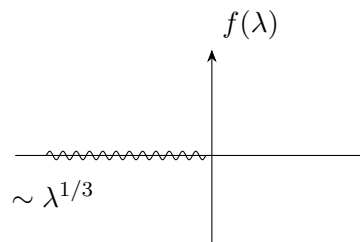
$$I(a) = \int_{-\infty}^{\infty} dx e^{-ax^2} = \sqrt{\frac{\pi}{a}} \sim a^{-1/2}$$

Branch cut starts at $a = 0$



$$\begin{aligned} I(a, \lambda) &= \int_{-\infty}^{\infty} dx e^{-ax^2 - \lambda x^4/4} \\ I_\lambda(a) &= \int_{-\infty}^{\infty} dx e^{-ax^2} e^{-\lambda x^4/4!} \\ &= \int_{-\infty}^{\infty} dx e^{-ax^2} \left(1 - \frac{\lambda x^4}{4!} + \frac{1}{2!} \frac{\lambda x^8}{4! 4!} - \dots \right) \\ I(a; m) &= \int_{-\infty}^{\infty} dx e^{-ax^2} x^m \\ \int_{-\infty}^{\infty} dx x^2 e^{-ax^2} &= -\frac{d}{da} I_0(a) \\ I_\lambda(a) &= I(a, 0) + \lambda I(a, 1) + \frac{\lambda^2}{2!} I(a, 2) + \dots \end{aligned}$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ \text{Not analytic} & \sim \frac{1}{\sqrt{a}} & \frac{1}{a^{5/2}} & \frac{1}{a^{7/2}} \\ \text{at } \lambda = 0 & & & \end{array}$$



Power series will only work iff
 λ is very small.
 Asymptotic series works at $\lambda = 0$.

HELING (arxiv)

↓

To understand renormalisation & fall in love with it. (could not find)

Renormalisation of λ :

$$\Gamma^{(4)}(k_1, k_2, k_3) =$$

Eliminate λ in favour of $g \equiv \Gamma^{(4)}(0, 0, 0)$

Replace μ^2 by m^2 , α^2 by a^2 ..

$$\Gamma^{(4)}(\{k\}) = \Gamma^{(4)}(\{k_i\}) - \Gamma^{(4)}(\{0\}) + g$$

Renormalisation condⁿ :

$$g = \lambda - \frac{\lambda^2}{2} \int \frac{d^d \tilde{q}}{(a^2 q^2 + m^2)} \quad (3 \text{ terms})$$

$$\left. \begin{array}{l} y = x - ax^2 \\ x = y - ay^2 \end{array} \right\} \quad (\text{for small } x)$$

$$\Gamma^{(4)}(\{k_i\}) = g + \frac{g^2}{2} \left(\int \cdots \right)$$

We choose $(k_1, k_2, k_3) = (0, 0, 0)$ as point to define ‘ g ’

we could have used other possibilities

$$g(k_1, k_2, k_3) = (k_1 \ k_2 \ k_3)$$

A common choice —

Symmetric pt. — all k_i are equal and have magnitude $\sqrt{\frac{3}{2}} k$

$$\Gamma^{(2)} = m^2 + k^2 \left(a^2 - \frac{g^2}{6} B(k, K) \right) + O(g^3)$$

$$\Gamma^{(4)} = g + \frac{g^2}{2} \left(\int () \cdots \right) + O(g^3)$$

$$\lambda = g \left(1 + \frac{3}{2} g \int \frac{d^d \tilde{q}}{(\alpha^2 q^2 + m^2)} \right) + O(g^3)$$

$$\mu^2 = m^2 - \frac{\lambda}{2} \int \frac{d^d \tilde{q}}{(\alpha^2 q^2 + m^2)} + \frac{\lambda^2}{6} A(0)$$

↘ substitute $g = g(\lambda)$

(Q) What about higher loops? What is minimum number of Λ -sensitive integrals?

We have 3 of them.

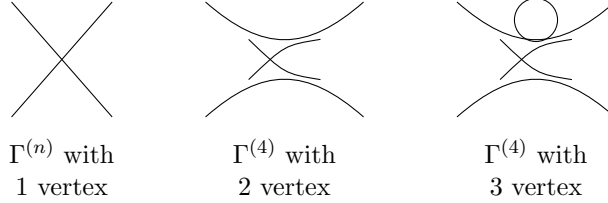
$$\begin{aligned}\mu^2 &= \mu^2(m^2, \lambda, \Lambda) \\ \lambda^2 &= \lambda^2(m^2, g, \Lambda) \\ \alpha^2 &= \alpha^2(m^2, \lambda, \Lambda)\end{aligned}$$

This is same as no. of bare parameters in L . So we have one parameter family of free parameters in L .

$$1 \text{ --- parameter family of lagrangians.} \\ (\alpha, \mu, \lambda, \Lambda)$$

all of which generates same renormalised theory, given by (a, m^2, g) .

Q) In general what is Λ -sensitivity for $\Gamma^{(n)}$?



Say $\Gamma^{(n)}$ has ' V ' vertices. Then internal lines

$$I = \frac{1}{2} (4V - n)$$

$$\text{No. of momenta} = n + \frac{1}{2} (4V - n)$$

$$\text{No. of individual momenta} = n + \frac{1}{2} (4V - n) - V$$

where $(n - 1)$ of these are independent external momenta.

$$(\text{No. of loop momenta} \equiv L) = V + \frac{n}{2} - (n - 1)$$

$$L = V - \frac{n}{2} + 1$$

In $\Gamma^{(n)}$ computation ; we will have integrals like

$$\frac{d^d \tilde{q}_1 d^d \tilde{q}_2 \cdots d^d \tilde{q}_L}{(\alpha^2 q_1^2 + \mu^2) (\alpha^2 q_2^2 + \mu^2) \cdots (\alpha^2 q_L^2 + \mu^2)}$$

goes like $q^{Ld-2L} \sim q^{(d-2)L} \sim q^{(d-2)(V-\frac{n}{2}+1)}$

look at large ' q ' behaviour —

For ex: $d = 4$ Behaviour of integral $\sim q^{2(V-\frac{n}{2}+1)}$

$$\sim q^{4-n} \quad \forall \text{ graphs of } \Gamma^{(n)}$$

$$\Gamma^{(2)} \sim q^2 \sim \Lambda^2$$

$$\Gamma^{(4)} \sim \ln \Lambda$$

$$\Gamma^{(n)} \sim (n > 4) \Rightarrow \text{not } \Lambda \text{ sensitive}$$

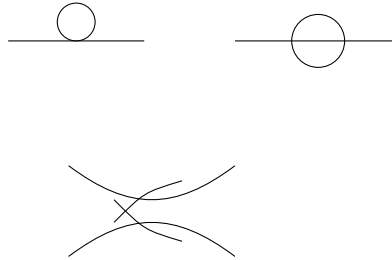
↘ Naive counting

1 P I (Particle irreducible)

These diagrams, cutting internal lines once will not be able to disconnect the diagrams / external lines.

A primitive divergent graph is one which is Λ -sensitive but has no Λ -sensitive ‘Sunset / sunrise / saturn’

only primitively
divergent
graphs.
($d \leq 4$)

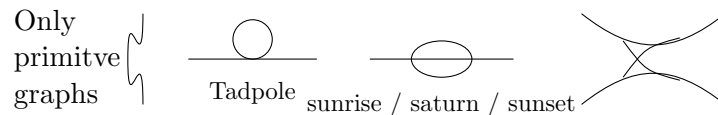


Every diagram ($\Gamma^{(n)}$; $n > 4$) is built out of $\Gamma^{(2)}$ & $\Gamma^{(4)}$.

9 Lecture 9

$$\text{Mid Term Exam} \begin{cases} 18^{\text{th}} \text{ or } 25^{\text{th}} \text{ Feb.} \\ 4 \text{ hours} \\ 100 \text{ points.} \end{cases}$$

We were talking about a graph that is Λ -sensitive has no Λ -sensitive subgraphs is called a primitively divergent graphs.



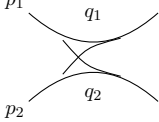
(in $\lambda\phi^4$ interaction, $d = 4$) (every other diagram is built out of $\Gamma^{(2)}$ & $\Gamma^{(4)}$)

Dimensional Regularisation — (D R)

$$\int_0^\Lambda \frac{q^3 dq}{q^2 + \mu^2} \sim \int_0^\Lambda \frac{q^3 dq}{q^2} \sim \int_0^\Lambda q dq \sim \Lambda^2$$

let's look at graph one by one.

1)  (Tadpole) : $\int \frac{d^4 \tilde{q}}{(q^2 + \mu^2)} \sim \Lambda^2$ quadratic UV-divergence

2)  : $\frac{\lambda^2}{2} \int \frac{d^4 \tilde{q}_1 d^4 \tilde{q}_2}{(q_1^2 + \mu^2)(q_2^2 + \mu^2)} \delta(q_1 + q_2 - p)$

$$\sim \frac{\lambda^2}{2} \int \frac{d^4 \tilde{q}}{(q_1^2 + \mu^2)((p - q_1)^2 + \mu^2)}$$

$$\longrightarrow \frac{\lambda^2}{2} \int \frac{d^4 \tilde{q}}{q^4} \sim \lambda^2 \ln \Lambda$$

If $\mu^2 = 0$; & $(p_1 + p_2)^2 = 0$, then there is also divergence for $q \rightarrow 0$ (IR divergence)

Typical integrals —

$$I_4(k) = \int d^4 \ell F(k, \ell)$$

For large $\ell \longrightarrow$ ℓ^{-2} for $\Gamma^{(2)}$
 ℓ^{-4} for $\Gamma^{(4)}$

For $d = 2$; $\int \frac{d^2 \ell}{\ell^4}$ is convergent in uv.

Suggest the following math. trick / procedure —

Consider, $I(w, k) = \int (d^{2w} \ell) F(\ell, k)$ ($2w = \text{dimension}$)

Treat 'w' as complex variable. (We are extending $\int d^d q$ integral to complex plane, via analytic continuation) $\overline{n+1} = n!$

We evaluate this integral in a region (of w-space), where there are no singularities. We then invent I' (new integral) which is convergent over bigger region / domain. This process is called analytic continuation in complex analysis.

$$\int d^{2w} \ell \boxed{F(\ell, k)} \quad \text{converges for } w < 2 \quad \left(\sim \frac{1}{\ell^2} \cdot \frac{1}{\ell^2} \right)$$

For our divergent integrals, this region is $w < 2$. We write :

$$\int d^{2w} \ell = \underbrace{\int d^4 \ell}_{\text{radial}} \underbrace{\int d^{2w-4} \ell}_{\text{angular}}$$

$$I = \int d^4\ell \underbrace{\int d\Omega_{2w-5}}_{\text{easy}} \int_0^\infty dL \frac{L^{2w-5}}{(L^2 + \ell^2 + \mu^2)}$$

$$\int_{-\infty}^\infty dx_1 \cdots \int_{-\infty}^\infty dx_N e^{-a(x_1^2 + x_2^2 + \cdots + x_N^2)} = \left(\frac{\pi}{a}\right)^{N/2}$$

Changing to polar coordinates

$$\underbrace{\int d\Omega_{N-1}}_{V(S^{N-1})} \int r^{N-1} dr e^{-ar^2} \quad (\sum x_i^2 = r^2)$$

↓ Volume of sphere of $(N-1)$ dim.

$$\Rightarrow V(S^{N-1}) \int r^{N-1} e^{-ar^2} dr \quad \begin{array}{l} ar^2 = t \\ ar dr = dt/2 \end{array}$$

$$V(S^{N-1}) \int \left(\frac{t}{a}\right)^{\frac{N-1}{2}} e^{-t} \frac{dt}{2a \left(\frac{t}{a}\right)^{1/2}}$$

$$\frac{1}{2a} V(S^{N-1}) \int \left(\frac{t}{a}\right)^{\frac{N-2}{2}} e^{-t} dt = \left(\frac{\pi}{a}\right)^{N/2}$$

$$\frac{1}{2a^{N/2}} V(S^{N-1}) \underbrace{\int t^{\frac{N}{2}-1} e^{-t} dt}$$

$$\frac{1}{2a^{N/2}} V(S^{N-1}) \Gamma(N/2) = \left(\frac{\pi}{a}\right)^{N/2}$$

$$\boxed{V(S^{N-1}) = \frac{2}{\Gamma(N/2)} \pi^{N/2}}$$

check: For $N=3$

$$V(S^2) = \frac{2\pi^{3/2}}{\Gamma(3/2)} = \frac{2\pi^{3/2}}{\frac{1}{2}\Gamma(1/2)} \quad (\Gamma(1/2) = \sqrt{\pi})$$

$$= 4\pi^{3/2-\frac{1}{2}} = 4\pi$$

So,

$$I = \frac{2\pi^{w-2}}{\Gamma(w-2)} \int d^4\ell \underbrace{\int_0^\infty \frac{dL L^{2w-5}}{(L^2 + \ell^2 + \mu^2)}}_{\text{easy}} \quad \begin{array}{l} 2w-6 \geq 0 \\ w \geq 3 \end{array}$$

uv divergent for $w \geq 3$

IR divergence for $w \leq 1$ (check by $\mu = 0$).

Write $L^{2w-5} = \frac{1}{w-2} \frac{d}{dL^2} \left((L^2)^{w-2} \right)$

$$dL L^{2w-5} = L dL L^{2w-6} = \frac{1}{2} dL^2 (L^2)^{w-3}$$

Integrate by parts over L^2 . & discard the surface term.

$$I = \underbrace{\frac{\pi^{w-2}}{(w-2)\Gamma(w-2)}}_{\Gamma(w-1)} \int d^4\ell \int dL^2 (L^2)^{w-2} \left(-\frac{d}{dL^2}\right) \frac{1}{L^4 + \ell^2 + \mu^2}$$

Again, IR divergent for $w \leq 1$
uv " " $w \geq 1$

Do integral by parts once more —

$$I = \frac{\pi^{w-2}}{\Gamma(w)} \int d^4\ell \int dL^2 (L^2)^{w-1} \left(-\frac{d}{dL^2}\right)^2 \frac{1}{L^4 + \ell^2 + \mu^2}$$

well defined for $(0 < w < 1)$

We need to look for I' which extends the domain & can be analytically continued to $w = 2$.

$$\text{use } 1 = \frac{1}{(5)} \left(\frac{\partial L}{\partial L} + \frac{\partial \ell_\mu}{\partial \ell_\mu} \right) \quad \left\{ \frac{\partial \ell_\mu}{\partial \ell_\mu} = 4 \right\}$$

Integrate by parts —

$$\begin{aligned} I &= -\frac{2\pi^{w-2}}{5\Gamma(w)} \int d^4\ell \int dL^2 \left(\ell_\mu \frac{\partial}{\partial \ell_\mu} + 2L^2 \frac{\partial}{\partial L^2} + 1 \right) \cdot \left(\frac{(L^2)^{w-1}}{(L^2 + \ell^2 + \mu^2)^3} \right) \\ &= \frac{-3\mu^2}{w-1} \frac{2\pi^{w-2}}{\Gamma(w)} \int d^4\ell \int_0^\infty dL^2 \frac{(L^2)^{w-1}}{(L^2 + \ell^2 + \mu^2)^4} \end{aligned}$$

Now, the only divergence is from the pole at $w = 2$.

Check,

$$\begin{aligned} I &= \int \frac{d^{2w}\ell}{(\ell^2 + \mu^2)^6} \longrightarrow I \\ I &= \frac{\pi^{w-2}}{\Gamma(w-1)} \int d^4\ell \int_0^\infty dL^2 (L^2)^{w-2} \left(-\frac{d}{dL}\right) \left(\frac{1}{L^4 + \ell^2 + \mu^2} \right) \\ I(w=2) &= \int d^4\ell \frac{(-1)}{(L^2 + \mu^2 + \mu^2)^6} \Big|_0^\infty = \int d^4\ell \frac{1}{(\ell^2 + \mu^2)^6} \end{aligned}$$

use,

$$\int \frac{d^{2w}\ell}{\ell^2 + \mu^2} = \frac{\pi^w \Gamma(1-w)}{\Gamma(1)} \cdot \frac{1}{(\mu^2)^{1-w}}$$

← expand around $w = 2$. (use $\epsilon = 2 - w$)

$$\Gamma(-n + \epsilon) = \frac{(-1)^n}{n!} \left(\frac{1}{\epsilon} + \psi(n+1) + \frac{\epsilon}{2} \left(\frac{\pi^2}{3} + \psi^2(n+1) - \psi'(n+1) \right) + O(\epsilon^2) \right)$$

$$\frac{d\psi(s)}{ds} = \frac{d}{ds} \ln \Gamma(s)$$

$$\psi(n+1) = 1 + \frac{1}{2} + \cdots - \frac{1}{n} - \gamma$$

↘ euler maclaurian const.

$$\psi(1) = -\gamma = -0.577$$

$$\psi'(n+1) = \frac{\pi^2}{6} - \sum_1^n \frac{1}{k^2} \quad \psi'(1) = \pi^2/6$$

So,

$$\lim_{w \rightarrow 2} \int \frac{d^{2w} \ell}{\ell^2 + \mu^2} = -\pi^2 \mu^2 \left(\frac{1}{2-w} + \psi(2) + O(\epsilon) \right)$$

↖ Blows up.

↘ quadratically divergent for $w = 2$

10 Lecture 10–11

(The scan of these notes is a single notebook marked “10, 11”. The break between the two lectures falls at “2-loop diagrams” below.)

ϕ^4 theory renormalisation —

$$S = \int d^{2w} x \left(\frac{1}{2} (\partial \phi)^2 - \frac{m^2 \phi^2}{2} - \frac{\lambda}{4!} \phi^4 (\mu^2)^{2-w} \right)$$

Here ‘ m ’ is bare mass in place of ‘ μ ’.

μ is introduced to make action dimensionless. & λ dimensionless in each ‘ $2w$ ’ dim.

(margin working)

$$\int d^{2w} x \left(-m^2 \phi^2 - \frac{\lambda}{4!} \phi^4 (\mu^2)^n \right) \quad [\varphi] = \frac{2w-2}{2} = w-1$$

$$1) \quad 2 + 2[\varphi] - 2w = 0$$

$$2) \quad -2w + 4[\varphi] + n(2) = 0$$

$$-2w + 4(w-1) + 2n = 0$$

$$2n = 2w - 4(w-1)$$

$$2n = -2w + 4$$

$$\underline{n = 2 - w}$$

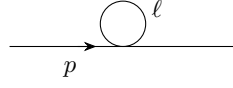
Minor changes in Feynman Rules :

1) Scalar product of k_1 & k_2 is summed over $2w$ components.

$$k_1 \cdot k_2 = k_1^0 k_2^0 - \left(\vec{k}_1 \cdot \vec{k}_2 \right)_{\text{in } (2w-1) \text{ dim}}$$

2) vertex $-\lambda \longrightarrow -\lambda (\mu^2)^{2-w}$

let's look at tadpole diagram —



$$T = \frac{1}{2} (-\lambda) (\mu^2)^{2-w} \int \frac{d^{2w} \ell}{(2\pi)^w} \frac{1}{(\ell^2 + m^2)} \quad \text{---(1)}$$

$\frac{1}{2}$: from where ?

(Digression) Integrals over arbitrary dimensions —

$$I_N = \int d^N \ell F(\ell)$$

goto polar coordinates — $(\ell_1, \ell_2 \dots \ell_N) \downarrow (L, \phi, \theta_1, \theta_2, \dots \theta_{N-2})$

$$L^2 = \ell_\mu \ell_\mu \quad (\text{Euclidean space})$$

$$d^N \ell = L^{N-1} dL (d\phi) (\sin \theta_1 d\theta_1) (\sin^2 \theta_2 d\theta_2) \dots (\sin^{N-2} \theta_{N-2} d\theta_{N-2})$$

$$0 \leq L < \infty, \quad 0 \leq \phi < 2\pi, \quad 0 \leq \theta_i < \pi$$

Standard Formulae —

$$\int_0^{\pi/2} (\sin t)^{2x-1} (\cos t)^{2y-1} dt = \frac{\Gamma(x) \Gamma(y)}{\Gamma(x+y)} \quad \begin{array}{l} \text{Re } x > 0 \\ \text{Re } y > 0 \end{array}$$

$\{A(2) \text{ of Ramond QFT}\}$

For $y = \frac{1}{2}$

$$\int_0^\pi \sin^k t dt = 2 \int_0^{\pi/2} \sin^k t dt = \frac{\sqrt{\pi} \Gamma(\frac{k+1}{2})}{\Gamma(\frac{k+2}{2})}$$

So,

$$I_N = \frac{\pi^{N/2}}{\Gamma(N/2)} \int_0^\infty dx x^{\frac{N-2}{2}} F(x) \quad (x = L^2) \quad \text{---(2)}$$

$$\left\{ \begin{array}{l} \text{comparing ① \& ②} \quad \frac{d^{2w} \ell}{(2\pi)^w} = dx \\ \frac{1}{\ell^2 + m^2} = x^{\frac{N-2}{2}} F(x) \end{array} \right.$$

For $F = (x+a)^{-A}$; $A = 2, 3, \dots$

$$\int_0^\infty dx \frac{x^{\frac{N-2}{2}}}{(x+a^2)^A} = (a^2)^{-A+\frac{N}{2}} \int_0^\infty dy y^{\frac{N-2}{2}} (1+y)^{-A}$$

$$\text{Recall, } B\left(\frac{N}{2}, A - \frac{N}{2}\right) = \frac{\Gamma(N/2) \Gamma(A - N/2)}{\Gamma(A)} = \int_0^\infty dy y^{\frac{N}{2}-1} (1+y)^{-A}$$

$$\begin{cases} \operatorname{Re}(N/2) > 0 \\ \operatorname{Re}(A - N/2) > 0 \end{cases}$$

$$\int \frac{d^N \ell}{(\ell^2 + a^2)^A} = \frac{\pi^{N/2} \Gamma(A - N/2)}{\Gamma(A)} \frac{1}{(a^2)^{A - N/2}}$$

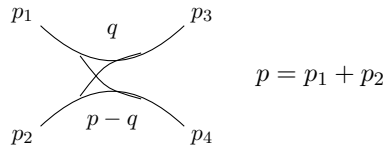
We take this formula valid to complex values of N by analytic continuation.

Another integral we need — (we are doing in euclidean signature)

$$\int d^N \ell \frac{\ell_\mu \ell_\nu}{(\ell^2 + 2p \cdot \ell + a^2)^A} = \frac{\pi^{N/2}}{\Gamma(A) (a^2 - p^2)^{A - \frac{N}{2}}} \left[\Gamma\left(A - \frac{N}{2}\right) p_\mu p_\nu + \frac{\delta_{\mu\nu}}{2} \Gamma\left(A - 1 - \frac{N}{2}\right) (a^2 - p^2) \right]$$

↓ Arbitrary vector.

Such Formula arrives in



Considering A can take complex values.

$$\begin{aligned} T &= \frac{1}{2} (-\lambda) (\mu^2)^{2-w} \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2)} \\ &= \frac{-\lambda m^2}{2 (4\pi)^2} \left(\frac{4\pi^2 \mu^2}{m^2} \right)^{2-w} \Gamma(1-w) \end{aligned}$$

expand it ① ↘ pole at $w = 1$.

(margin working)

$$\begin{aligned} \Gamma(m) &= (m-1)! = \int_0^\infty t^{m-1} e^{-t} dt \\ \Gamma(1-w) &= (1-w-1)! = \int_0^\infty t^{-w} e^{-t} dt \\ &= \int_0^\infty t^{-w} \left(1 - t + \frac{t^2}{2!} - \frac{t^3}{3!} + \dots \right) dt \\ &= \int_0^\infty dt \left(t^{-w} - t^{1-w} + \frac{t^{2-w}}{2!} + \dots \right) = \left(\frac{t^{1-w}}{-w+1} - \frac{t^{1-w}}{2-w} + \frac{t^{2-w}}{(3-w)2!} + \dots \right) \Big|_0^\infty \end{aligned}$$

Expand about $2-w$ ($\equiv \epsilon$)

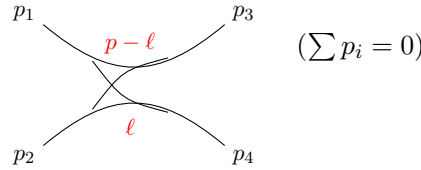
$$T = \frac{-\lambda m^2}{32\pi^2} \left[1 + (2-w) \ln \left(\frac{4\pi\mu^2}{m^2} \right) + \dots \right] \left[\frac{-1}{2-w} - \psi(2) + \dots \right]$$

pole term from lowest series $\downarrow$

$$= \frac{\lambda m^2}{32\pi^2} \left(\frac{1}{2-w} + \psi(2) - \underbrace{\ln \left(\frac{m^2}{4\pi^2 \mu^2} \right)}_{\text{finite part}} + O(2-w) \right)$$

- The divergence in $T \sim$ simple pole (in ϵ)
- The finite part is completely arbitrary. (depends on μ^2) as different people have diff. values of μ .

Next look at $\Gamma^{(4)}$



$$= \frac{+1}{2} (-\lambda)^2 (\mu^2)^{4-2w} \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2)} \frac{1}{(\ell - p)^2 + m^2}$$

$$(P = p_1 + p_2)$$

Feynman parameterization —

$$\frac{1}{D_1^{a_1} D_2^{a_2} \cdots D_K^{a_K}} = \frac{\Gamma(a_1 + a_2 + \cdots + a_K)}{\Gamma(a_1) \Gamma(a_2) \cdots \Gamma(a_K)} \int_0^1 \cdots \int_0^1 dx_1 dx_2 \cdots dx_K$$

$$\times \frac{\delta(1 - (x_1 + x_2 + \cdots + x_K))}{(D_1 x_1 + D_2 x_2 + \cdots + D_K x_K)^{(a_1 + a_2 + \cdots + a_K)}} \left(x_1^{a_1-1} x_2^{a_2-1} \cdots x_K^{a_K-1} \right)$$

$$\frac{1}{(\ell^2 + m^2) ((\ell + p)^2 + m^2)} = \int_0^1 dx \frac{1}{[\ell^2 + m^2 + 2\ell \cdot p(1-x) + p^2(1-x)]^2}$$

$$\downarrow (\ell - p)^2 \quad (?)$$

$$D_r = (\ell')^2 + m^2 + p^2 x(1-x);$$

$$d^{2w} \ell' = d^{2w} \ell$$

So,  is $= \frac{\lambda^2}{2} (\mu^2)^{4-2w} \int_0^1 dx \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2 + p^2 x(1-x))^2}$

use,

$$\int \frac{d^{2w} \ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + M^2 + 2\ell \cdot p)^A} = \frac{\Gamma(A-w)}{(4\pi)^w \Gamma(A)} \frac{1}{(M^2 - p^2)^{A-w}}$$

$$(\text{bubble}) = \frac{\lambda^2}{2} (\mu^2)^{4-2w} \int_0^1 dx \frac{\Gamma(2-w)}{(4\pi)^w} \frac{1}{(M^2 + p^2 x(1-x))^{2-w}}$$

Expand in powers of ϵ ($= 2 - w$) :

$$(\text{bubble}) = (\mu^2)^{2-w} \frac{\lambda}{32\pi^2} \left(\frac{1}{2-w} + \psi(1) - \int_0^1 dx \ln \left(\frac{M^2 + p^2 x(1-x)}{4\pi\mu^2} \right) \right)$$

The finite part depends upon μ^2 & p (ext. momentum)

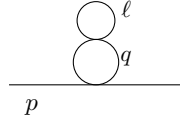
We know

$$\int_0^1 dx \ln \left(1 + \frac{4x}{a}(1-x) \right) = -2 + \sqrt{1+a} \ln \left(\frac{\sqrt{1+a}+1}{\sqrt{1+a}-1} \right)$$

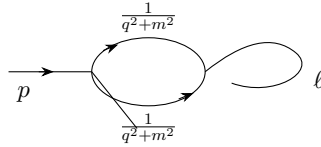
Using above,

$$\begin{aligned} (\text{bubble}) &= (\mu^2)^{2-w} \left(\frac{\lambda}{32\pi^2} \right) \left[\frac{1}{2-w} + \psi(1) + 2 + \ln \frac{4\pi\mu^2}{m^2} \right. \\ &\quad \left. - \left(1 + \frac{4m^2}{p^2} \right)^{1/2} \ln \left(\frac{\sqrt{1 + \frac{4m^2}{p^2}} + 1}{\sqrt{1 + \frac{4m^2}{p^2}} - 1} \right) \right] (\ln 4\pi m) \end{aligned}$$

2-loop diagrams — To be done!



$$= \frac{\lambda^2}{4} (\mu^2)^{4-2w} \int \frac{d^{2w}\ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2)} \int \frac{d^{2w}q}{(q^2 + m^2)(q^2 + m^2)}$$



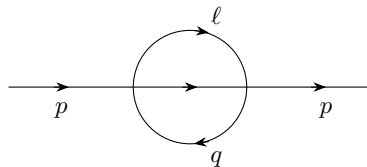
Subleading part $\downarrow$

$$= \frac{\lambda^2 m^2}{1024 \pi^4} \left(\frac{1}{(2-w)^2} + \frac{1}{(2-w)} \left(2 \ln \frac{4\pi\mu^2}{m^2} + \psi(2) + \psi(1) \right) \right)$$

↙ arbitrary

$$\begin{aligned} &+ 2 \ln^2 \frac{4\pi\mu^2}{m^2} + 2 \ln \frac{4\pi\mu^2}{m^2} (\psi(2) + \psi(1)) \\ &+ \frac{1}{2} \left\{ (\psi(2) + \psi(1))^2 + \frac{2\pi^2}{3} - \psi'(2) - \psi'(1) + O(2-w) \right\} \end{aligned}$$

The other 2-loop diagram —



$$\Sigma(p) = \frac{\lambda^2}{6} (\mu^2)^{4-2w} \int \left\{ \frac{d^{2w} \hat{\ell}}{(2\pi)^{2w}} \frac{d^{2w} \check{q}}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2)} \frac{1}{(q^2 + m^2)} \frac{1}{((P - \ell + q)^2 + m^2)} \right\}$$

$$\text{use} \quad \frac{1}{4w} \left(\frac{\partial \ell_\mu}{\partial \ell_\mu} + \frac{\partial q_\mu}{\partial q_\mu} \right) = 1 \quad \left(\begin{array}{l} \downarrow 2w \\ \downarrow 2w \end{array} \right)$$

(margin working)

$$\begin{aligned} \Sigma(p) &= \frac{\lambda^2}{6} (\mu^2)^{4-2w} \int \left\{ \frac{d^{2w} \hat{\ell}}{(2\pi)^{2w}} \frac{d^{2w} \check{q}}{(2\pi)^{2w}} \cdot \left(\ell_\mu \frac{\partial}{\partial \ell_\mu} + q_\mu \frac{\partial}{\partial q_\mu} \right) \right. \\ &\quad \times \frac{1}{(\ell^2 + m^2)} \frac{1}{(q^2 + m^2)} \frac{1}{((p + q - \ell)^2 + m^2)} \end{aligned}$$

Integrate by parts

$$\begin{aligned} \Sigma(p) &= \frac{1}{4w} \frac{\lambda^2}{6} (\mu^2)^{4-2w} \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \int \frac{d^{2w} q}{(2\pi)^{2w}} \\ &\quad \times \frac{3m^2 + p \cdot (p + q - \ell)}{(q^2 + m^2) (\ell^2 + m^2) [(q + p - \ell)^2 + m^2]^2} \\ &= \frac{1}{2w-3} \frac{\lambda^2}{6} (\mu^2)^{4-2w} [3m^2 K(p) + p^\mu K_\mu(p)], \text{ where} \\ K(p) &= \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \int \frac{d^{2w} q}{(2\pi)^{2w}} \frac{1}{(q^2 + m^2) (\ell^2 + m^2) ((q + p - \ell)^2 + m^2)} \\ K_\mu(p) &= \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \int \frac{d^{2w} q}{(2\pi)^{2w}} \frac{(p + q - \ell)_\mu}{(q^2 + m^2) (\ell^2 + m^2) ((q + \ell - p)^2 + m^2)} \end{aligned}$$

Using Feynman parameterization —

$$K(p) = \int d^{2w} \tilde{\ell} d^{2w} \tilde{q} \frac{1}{(q^2 + m^2)} \int_0^1 dx \frac{1}{\left(\underbrace{\ell^2 + m^2 + (p + q)^2 x(1-x)}_{M^2} \right)^2}$$

do the ℓ -integral first

$$\int d^{2w} \ell \frac{1}{(\ell^2 + M^2 + 2\ell \cdot p)^A} = \frac{\Gamma(A-w)}{(4\pi)^w \Gamma(A)} \frac{1}{(M^2 - p^2)^{A-w}}$$

Comparing $A=2$ ($\Gamma(2) = 1! = 1$)

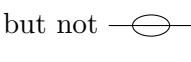
$$K(p) = \frac{\Gamma(2-w)}{(4\pi)^w} \int_0^1 dx \int d^{2w} \check{q} \frac{1}{(q^2 + m^2)^2} \frac{1}{[m^2 + (p + q)^2 x(1-x)]^{2-w}}$$

use Feynman parameterization once more to deal with

$$\frac{1}{(q^2 + m^2)^2} \frac{1}{((p + q)^2 x(1-x) + m^2)^{2-w}}$$

$$K(p) = \frac{\Gamma(2-w)}{(4\pi)^w} \int_0^1 dx (x(1-x))^{w-2} \int_0^1 dy y^{1-w}(1-y) \cdot \int \frac{d^{2w}q}{(2\pi)^w} (q^2 + p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right))^{w-4}$$

- DR allows us to do more than one loop calculations.

- In φ^3 theory counterterm kills  but not  & .

do the q integral first,

$$K(p) = \frac{\Gamma(4-2w)}{(4\pi)^{2w}} \int_0^1 dx (x(1-x))^{w-2} \int_0^1 dy y^{1-w}(1-y) \times \left(p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right)^{-2\epsilon}$$

Expand around $\epsilon = 0$ ($w = 2$).

We have to isolate poles from finite part.

$$\begin{aligned} K(p) &= \frac{\Gamma(2\epsilon)}{(4\pi)^{4-2\epsilon}} \int_0^1 dx (x(1-x))^{-\epsilon} \int_0^1 dy y^{-1+\epsilon}(1-y) \\ &\times \left[p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right]^{-2\epsilon} \\ &= \text{use } y^{-1+\epsilon} = \frac{1}{\epsilon} \frac{d}{dy} y^\epsilon, \text{ integrate by parts} \\ K(p) &= \frac{\Gamma(2\epsilon)}{(4\pi)^{4-2\epsilon}} \frac{1}{\epsilon} \int dx (x(1-x))^{1-\epsilon} \int_0^1 dy y^\epsilon \times \left\{ 1 + 2\epsilon(1-y) \frac{d}{dy} \right. \\ &\times \ln \left[p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right] \\ &\times \left. p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right\}^{-2\epsilon} \end{aligned}$$

Expand the integral about $\epsilon = 0$

$$K(p) = \frac{\Gamma(2\epsilon)}{(4\pi)^{4-2\epsilon}} \frac{1}{\epsilon} (1 + \epsilon - 2\epsilon \ln m^2 + O(\epsilon^2))$$

Now, Evaluate $K_\mu(p)$ similarly

{Mathew's & walker metn}

Since we are working in Euclidean metric. ($K_\mu = K^\mu$).

$$\begin{aligned} p^\mu K_\mu(p) &= \frac{p^2 \Gamma(2\epsilon)}{(4\pi)^{4-2w}} \int_0^1 dx (x(1-x))^{1-\epsilon} \int_0^1 dy y^\epsilon (1-y) \\ &\times \left[p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right]^{-2\epsilon} \end{aligned}$$

Again expand around $\epsilon = 0$.

$$p^\mu K_\mu(p) = \frac{p^2 \Gamma(2\epsilon)}{(4\pi)^{4-2\epsilon}} \left(\frac{1}{2} + O(\epsilon) \right)$$

Putting All together :

$$\Sigma(p) = \frac{-\lambda^2}{6(16\pi^2)^2} \left[\frac{3m^2}{2\epsilon^2} + \frac{3m^2}{\epsilon} \left(\frac{1}{2} + \psi(1) + \ln \frac{4\pi^2 \mu^2}{m^2} \right) + \frac{p^2}{4\epsilon} + \text{finite term} \right]$$

Renormalisation —

poles in ‘ ϵ ’ can be eliminated order by order in λ .

$$\left(\hat{\lambda} \equiv \frac{\lambda}{16\pi^2} \right), \quad \hat{m}^2 = \frac{m^2}{16\pi^2 \mu^2}$$

Look at the tadpole digram —

$$\begin{array}{c} \text{---} \bigcirc \text{---} \\ \\ = m^2 \frac{\hat{\lambda}}{2} \left(\frac{1}{\epsilon} + \psi(2) - \ln \hat{m}^2 + O(\epsilon) \right) \quad \text{---} \textcircled{3} \end{array}$$

$\nwarrow$ divergent as $\epsilon \rightarrow 0$

finite $F_1(\epsilon, m^2)$

Original lagrangian

$$\mathcal{L} = \frac{1}{2} (\partial\varphi)^2 - \frac{m^2 \varphi^2}{2} - \frac{\lambda}{4} \varphi^4 - \underbrace{g \varphi^2}_{\mathcal{L}_{int}}$$

$$\begin{array}{c} \text{bare mass} \\ \longleftarrow \quad \frac{\times}{g} \end{array}$$

(Choose coff g such that Counter term kills off divergence in $\text{---} \bigcirc \text{---}$ digram.)

Actually, Bare $\mathcal{L}$ itself is not defined as $\epsilon \rightarrow 0$ bare parameters depend on ϵ .

Add an extra term

$$\frac{m^2}{4} \hat{\lambda} \left(\frac{1}{\epsilon} + F_1(\epsilon, m^2) \right) \varphi^2$$

, where F_1 is arbitrary (dimensionless) function, analytic as $\epsilon \rightarrow 0$.

Puzzle :

Finite part is dependent on renormalization scheme, but we expect it to be same, wheather we do cut off renormalisation or DR. So how to reconcile ?

Our $\mathcal{L}$ looks like :

$$\mathcal{L} = \frac{1}{2} (\partial\varphi)^2 - \frac{m^2 \varphi^2}{2} - \frac{\lambda}{4!} \varphi^4 + \underbrace{\frac{m^2 \hat{\lambda}}{4} \left(\frac{1}{\epsilon} + F_1 \right) \varphi^2}_{\text{Tree level contribution}}$$

$\downarrow$ Tree level contribution

This $\mathcal{L}$ has extra Feynman rule —

$$\text{---}\times\text{---} = -\frac{1}{2} m^2 \hat{\lambda} \left(\frac{1}{\epsilon} + F_1 \right) \quad \text{---(4)}$$

Lets compute $\Gamma^{(2)}$ or inverse propagator —

$$\Gamma_{\text{new}}^{(2)} = \text{---} + \text{---}\bigcirc\text{---} + \text{---}\times\text{---} + O(\lambda)$$

eq ③ ↓

$$\begin{aligned} &= p^2 + m^2 + \frac{m^2 \hat{\lambda}}{2} \left[\frac{1}{\epsilon} + \psi(2) - \ln \hat{m}^2 + O(\epsilon) \right] \\ &\quad - \frac{1}{2} m^2 \hat{\lambda} \left(\frac{1}{\epsilon} + F_1 \right) + O(\hat{\lambda}) \\ &= p^2 + m^2 + \frac{m^2 \hat{\lambda}}{2} (\psi(2) - \ln \hat{m}^2 - F_1) + O(\hat{\lambda}) \\ &= p^2 + m^2 \left(1 + \frac{\hat{\lambda}}{2} (\psi(2) - \ln \hat{m}^2 - F_1) \right) + O(\hat{\lambda}) \end{aligned}$$

→ means upto one loop.

2 pt. Fun. is now finite (upto order $\hat{\lambda}$)

The extra term is called counterterm $\left(\frac{m^2}{4} \hat{\lambda} \left(\frac{1}{\epsilon} + F_1 \right) \right)$

look at $O(\lambda^2)$

The 1 P I 4 pt. function —

$$\begin{array}{c} \text{---}\bigcirc\text{---} \\ \downarrow \\ \text{Not 1PI } \textcolor{red}{?} \end{array} = \text{---}\times\text{---} + \text{---}\times\text{---} + \text{---}\times\text{---} + \text{---}\times\text{---} + O(\lambda^3)$$

$$\begin{aligned} \Gamma^{(4)}(p_1, p_2, p_3, p_4) &= -\mu^{2\epsilon} \lambda \left(1 - \frac{3\hat{\lambda}}{2} \left(\frac{1}{\epsilon} + \psi(1) + 2 + \ln \hat{m}^2 \right. \right. \\ &\quad \left. \left. - \frac{1}{3} A(s, t, u) + O(\epsilon) \right) \right) + O(\lambda^3) \end{aligned}$$

$$A(s, t, u) = \sum_{z=s,t,u} \left(1 + \frac{4m^2}{z} \right)^{1/2} \ln \left(\frac{\left(1 + \frac{4m^2}{z} \right)^{1/2} + 1}{\left(1 + \frac{4m^2}{z} \right)^{1/2} - 1} \right)$$

s, t, u are mendelstam variables

$$s = (p_1 + p_2)^2 = (p_1 + p_2) \cdot (p_1 + p_2)$$

$$t = (p_1 + p_3)^2$$

$$u = (p_1 + p_4)^2$$

$$\left(\Gamma^{(4)} \text{ diverges as } \frac{1}{\epsilon} \right)$$

So we add a counterterm to $\mathcal{L}$ of the form —

$$\text{Counterterm} = \frac{1}{4} \mu^{2\epsilon} \lambda \frac{3\hat{\lambda}}{2} \left(\frac{1}{\epsilon} + G_1(\epsilon, m^2) \right) \varphi^4$$

↓ arbitrary but analytic as $\epsilon \rightarrow 0$.

One more Feynman rule gets to be added due to counterterm.

$$\text{Diagram: four lines meeting at a central point} = -\frac{3}{2} \mu^{2\epsilon} \lambda \hat{\lambda} \left(\frac{1}{\epsilon} + G_1 \right)$$

$$\Gamma_{\text{(new)}}^4 = \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} + \text{Diagram 4} + \text{Diagram 5}$$

$$\begin{aligned} \Gamma_{\text{new}}^{(4)}(p_1, p_2, p_3, p_4) = & -\mu^{2\epsilon} \lambda \left(1 - \frac{3}{2} \hat{\lambda} (-G_1 + \psi(1) + 2 + \ln \hat{m}^2 \right. \\ & \left. - \frac{1}{3} A(s, t, u) \right) + O(\epsilon) + O(\lambda^3). \end{aligned}$$

Now divergence is gone!

We need only to renormalize $\Gamma^{(2)}$ & $\Gamma^{(4)}$ then all 8 point & 16 point functions